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Thesis for the Degree of Master

**A Study on the Etching
Characteristics of High Aspect
Ratio Oxide Etching using C_4F_6
Plasma in Inductively Coupled
Plasma with Low Frequency Bias
Power**

by

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Department of Control and
Instrumentation Engineering
Graduate School
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May. 2024

權 光 虎 教 授 指 導
碩 士 學 位 論 文

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이 論文을 工學 碩士學位論文으로 出함.

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Abstract

A Study on the Etching Characteristics of High Aspect Ratio Oxide Etching using C_4F_6 Plasma in Inductively Coupled Plasma with Low Frequency Bias Power

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In this thesis, a study on high aspect ratio (HAR) oxide etching processes in inductively coupled plasma (ICP) using C_4F_6 gas and low-frequency (2 MHz) bias power was conducted. Amid growing efforts to reduce carbon emissions in the semiconductor industry, C_4F_6 is environmentally friendly with a Global Warming Potential (GWP) of less than 1. Thus, to verify the feasibility of using C_4F_6 for HAR oxide etching, we investigated the etching characteristics of $CF_4 + X + He$ mixed gas ($X = C_4F_6$) on Amorphous Carbon



Layer (ACL) and Silicon Oxide (SiO_2), including etch rates, etch selectivity, and etch profiles of the X gas. Optical Emission Spectroscopy (OES) and a dual Langmuir probe were employed for plasma diagnostics. Compared to conventional C_4F_8 mixed plasma, the C_4F_6 mixed plasma demonstrated similar SiO_2 /ACL selectivity and superior etch profiles. The chemical changes on the etched ACL and SiO_2 surfaces were analyzed using X-ray Photoelectron Spectroscopy (XPS), supporting the above results by showing increased passivation due to the rise in CF_x species. Additionally, experiments were conducted to study the correlation of etching characteristics with variations in source power and bias power. Plasma diagnostics revealed that as the source power increased, the ion density and electron temperature contributing to plasma generation also increased. With an increase in bias power, the value of $-\text{V}_{\text{dc}}$ associated with ion bombardment energy



increased. Furthermore, the etch profiles derived from these parameters highlighted the etching characteristics corresponding to each variable. These etching results indicate that the use of C_4F_6 gas can significantly contribute to the development of next-generation HAR oxide etching processes.



요 약

저주파 바이어스 유도결합 플라즈마에서 C_4F_6

플라즈마를 이용한 고중형비 산화물 에칭의 에칭

특성에 관한 연구

성 명: 우 병 준

학 과: 제어계측공학과

전 공: 계측 및 센서

지도교수: 권 광 호 교수님

본 논문에서는 C_4F_6 gas 로 저주파(2 MHz) 바이어스 파워를 이용하여 유도 결합 플라즈마에서의 고 중형비(HAR) 산화물 식각 공정 연구를 진행하였습니다. 반도체 산업에서 탄소저감을 위한 노력이 주목받는 가운데 C_4F_6 는 GWP <1 로 환경 친화적입니다. 따라서 C_4F_6 를 이용한 고 중형비 산화막 에칭 가능성을 확인하기 위하여 $CF_4 + X + He$ 혼합 가스 ($X=C_4F_6$)를 사용하여 Amorphous Carbon Layer(ACL) 및 Silicon



Oxide (SiO_2)에 대한 식각 속도, 식각 선택도 및 식각 프로파일과 같은 X gas의 식각 특성을 조사했습니다. 플라즈마 진단을 위해 광학 방출 분광법과 이중 랭뮤어 프로브 측정을 수행했습니다. 기존의 C_4F_8 혼합 플라즈마와 비교하여 C_4F_6 혼합 플라즈마는 SiO_2/ACL 선택 비 및 식각 프로파일이 유사하고, 식각 프로파일도 우수하였습니다. X선 광전자 분광법을 사용하여 식각된 ACL, SiO_2 표면의 화학적 변화를 분석하여 CFX 중의 증가로 인한 passivation의 증가로 인한 것으로 위의 결과를 뒷받침 하였습니다. 또한 소스 전력 변화, 바이어스 전력 변화에 따른 실험을 통해 각각의 파라미터에 대한 식각 특성과의 상관관계를 연구하였습니다. 플라즈마 진단을 통해 소스 전력이 증가함에 따라 플라즈마 생성에 기여하는 이온밀도 및 전자온도가 증가하는 것을 확인하였고, 바이어스 전력이 증가함에 따라 ion bombardment energy 와 연관된 $-V_{dc}$ 의 값이 증가하는 것을 확인했습니다. 또한 각각의 식각 프로파일 도출을 통해 변수에



따른 식각 특성을 도출했습니다. 이러한 식각 결과는 또한 C_4F_6 가스의 사용이 차세대 HAR 산화물 식각 개발에 크게 기여할 수 있음을 나타냅니다.



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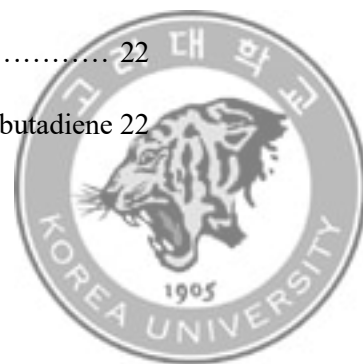


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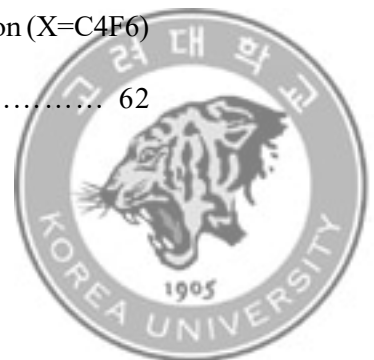


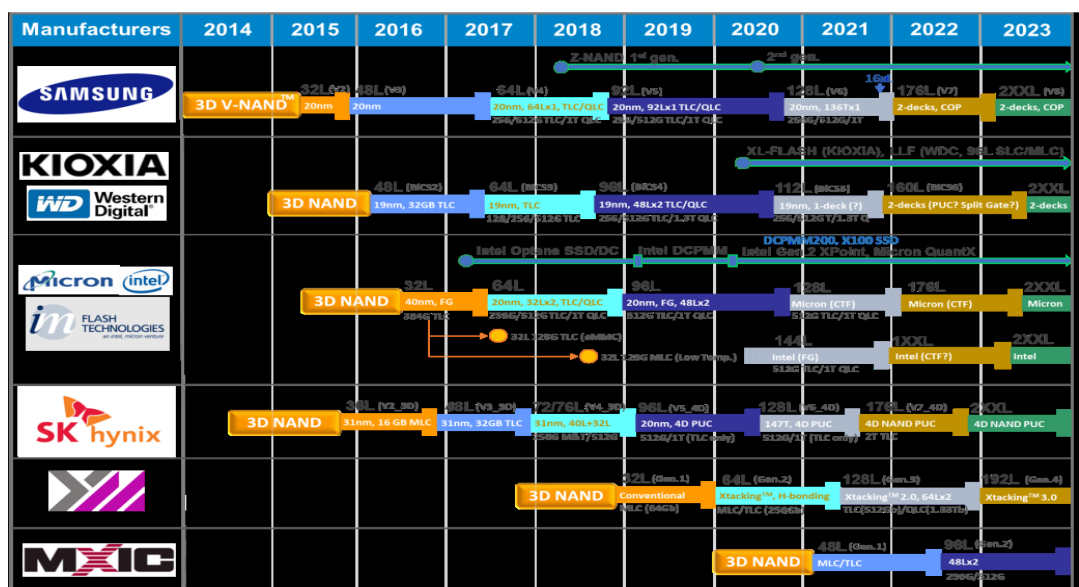
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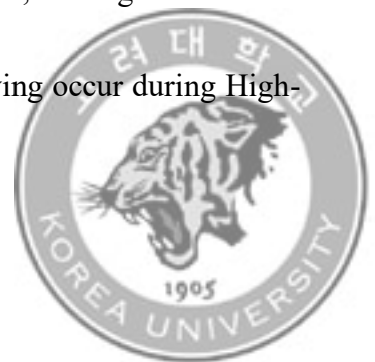


Introduction



Currently, the semiconductor industry is reaching the limits of miniaturization with the advancement of microfabrication technologies for high performance, low power consumption, and miniaturization. This has led to increased interest in alternative

technologies such as advanced packaging. Enhancing chip performance and production efficiency through advanced processes for pre-processing and post-processing, especially achieving integration below 3nm and 3D integration, is a primary focus. Therefore, the development of microprocessors for detailed circuit implementation is essential. In particular, the transition from 2D planar structures to 3D vertical stacking structures in NAND architecture has made the formation of nanoscale patterns in semiconductor devices dependent on five key processes: deposition, lithography, etching, ion implantation, and planarization. Among these processes, etching is gaining more attention in semiconductor manufacturing because it is repeatedly applied alongside deposition and lithography in increasingly complex structures. If the desired area is not etched cleanly during etching, misalignment can occur during multi-patterning processes due to uneven etch profiles, referred to as misalignment caused by rough etch surfaces. [1] During etching, the ion incident angle is influenced by the mask shape, making it difficult to obtain vertical profiles, and issues such as necking and bowing occur during High-



Aspect-Ratio (HAR) etching processes. [2] Therefore, optimizing the etching process through appropriate control of etching gases is essential to adjust the mask shape and ensure the formation of vertical profiles.

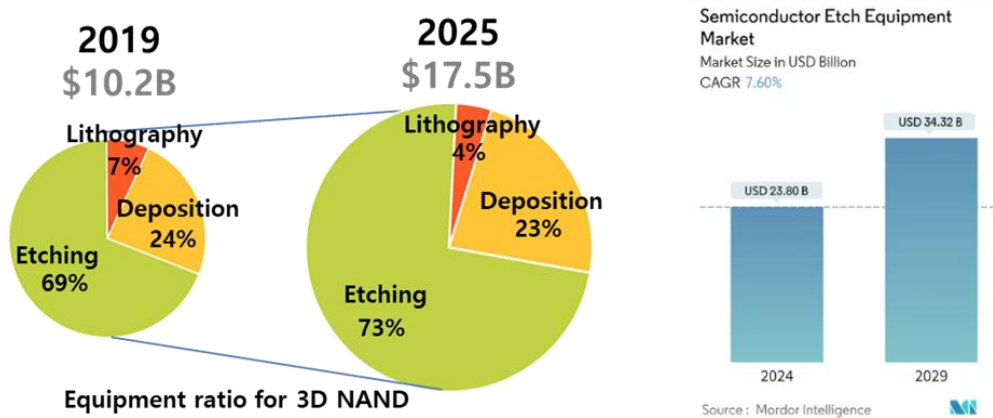


Figure 1.2. Estimated share of etching process in global market



1.2 Dry-etching technology

Etching can be divided into dry etching and wet etching. Wet etching involves the use of reactive solutions to selectively remove materials. However, wet etching results in isotropic etching, where the etch rate is high and uniform in both vertical and horizontal directions. Additionally, the pattern size achievable is limited, making it impractical for smaller sizes. On the other hand, plasma etching selectively removes materials using reactive gases, vapors, or ions generated by plasma. In current semiconductor processes, as the component sizes decrease, and circuits become more complex, anisotropic etching in dry etching processes is essential. Plasma dry etching overcomes the issue of isotropy and provides higher patterning resolution. However, due to the complexity of gas reactions and etching mechanisms within the plasma, as well as various other parameters, modeling and controlling plasma etching can be challenging.



1.2.1 isotropic etching and anisotropic etching

When etching progresses at the same rate in both the lateral (x-direction) and downward (y-direction) directions, it is called isotropic etching, resulting in etching similar to that shown in Figure 1-(a). Conversely, when the etching rate in the y-direction is greater than that in the x-direction, it is called anisotropic etching, as shown in Figure 1-(b). In particular, when the etching rate in the x-direction is zero, as depicted in Figure 1-(c), it represents an extreme case of anisotropic etching known as vertical etching. Directional etching, as shown in Figure 1-(c), occurs when the etching rate in the x-direction is lower than the etching rate in the y-direction, and this type of etching is generally achievable in plasma etching.

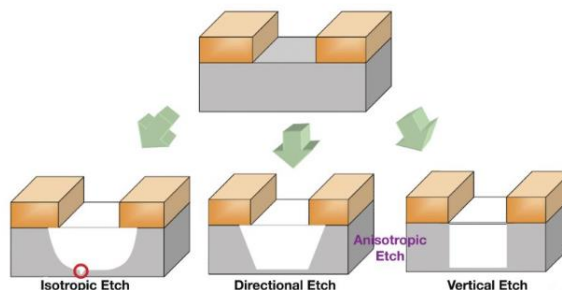


Figure 1.3 isotropic etching and anisotropic etching



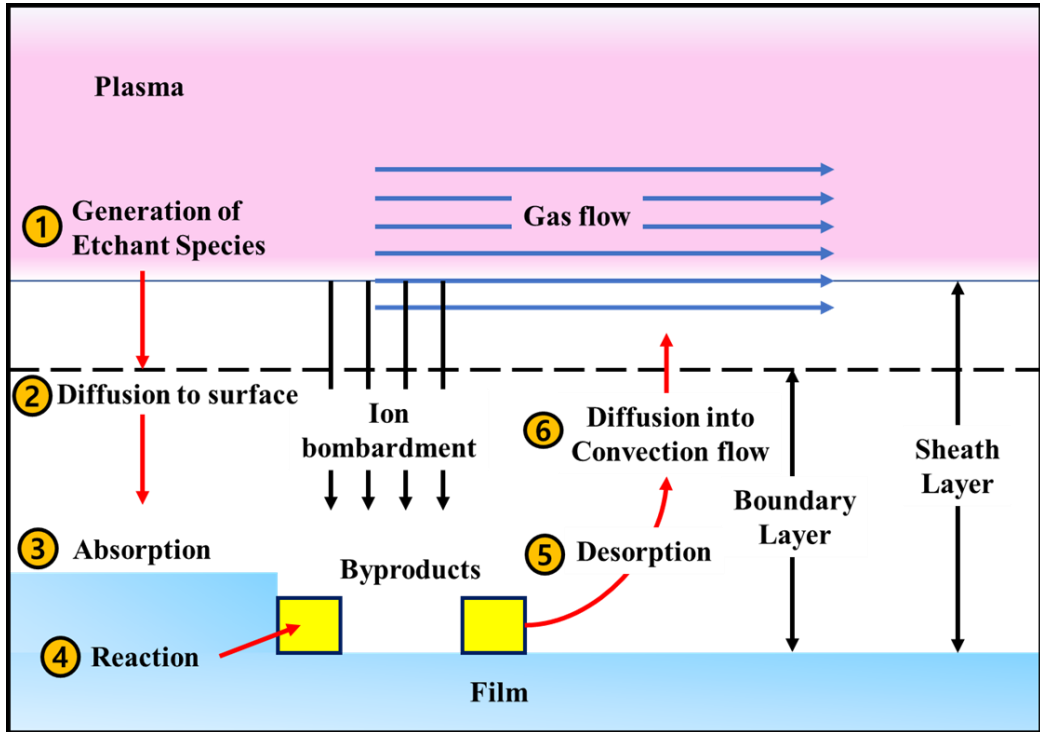
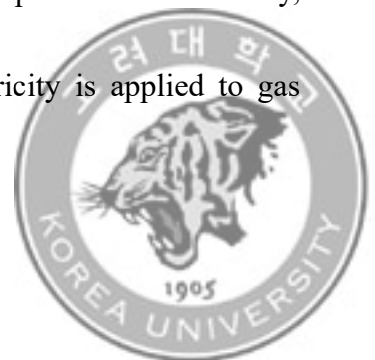


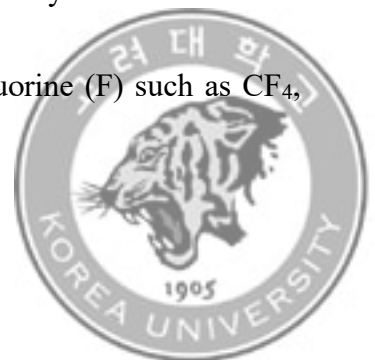
Figure 1.4. Plasma etching schematic diagram in the process of plasma dry etching,

In plasma dry etching processes, reactive gases are introduced into a vacuum chamber, and a strong electric field is applied by a power source. This accelerates electrons, which then collide with neutral atoms, leading to ionization, recombination, dissociation, abstraction, and fragmentation processes. Ultimately, plasma refers to ionized gas. When energy such as electricity is applied to gas



composed of atoms and molecules, plasma is formed, consisting of electrons, ions, dissociated gases, and photons, and it becomes electrically neutral. The etching species in the plasma then diffuse towards the target surface, where they are adsorbed. Due to the voltage applied to the electrode, ions gain directionality, and their impact weakens the molecular bonds. The etching species adsorbed at the weakened bond sites react more easily with the target material, forming volatile compounds. These compounds then desorb from the surface, diffuse into the gas phase, and are eventually evacuated by vacuum equipment. Dry etching technology utilizes chemical reactions between surface materials and the impacts of ions to facilitate precise pattern transfer for nanoscale patterning processes. [3]

A representative material for silicon insulators is silicon dioxide (SiO_2). To etch insulators, it is necessary to use etching reactive gases that have good reactivity with each material and form volatile by-products. In today's semiconductor manufacturing industry, gases with a high proportion of fluorine (F) such as CF_4 ,

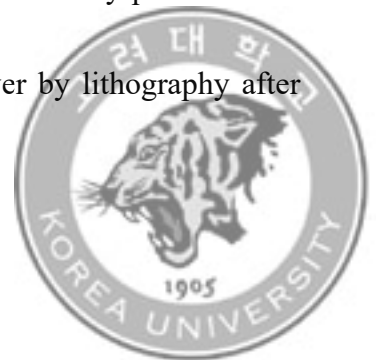


C_2F_6 , C_4F_6 , and C_4F_8 , which induce dry etching on the substrate surface due to their high reactivity with F, are commonly used. [4-9] Additionally, various studies focus on adding oxygen and adjusting the gas ratio to control the fluorocarbon (FC) polymer layer, achieving anisotropic etch profiles through appropriate masking and sidewall passivation.



1.3 Multiple-patterning technology

For decades, miniaturization has been a major trend in the electronics market, leading to the integration of more functions into single products and technological advancements. Achieving these required improvements in existing lithography, which led to the adoption of multiple patterning technology. Multiple patterning includes overlaying smaller and denser patterns onto patterns engraved over a wide area. Double patterning methods are broadly classified into SADP (Self-Aligned Double Patterning) and LELE (Litho-Etch-Litho-Etch), which are extensively used in the NAND flash and RAM memory manufacturing industries [10,11]. Among them, one of the most common double patterning techniques is the LELE method, which alternates lithography and etching steps twice. Particularly, the flexibility in design is greatly enhanced as each mask can be independently optimized. In this process, a perfect photomask pattern is divided into two low-density patterns. The first pattern is transferred to the underlying hard mask layer by lithography after



etching, and this process is repeated for the second pattern. The key to this process is that the two patterns must be perfectly aligned. Imperfect alignment causes the combined patterns to have significant relative displacements, leading to yield issues.

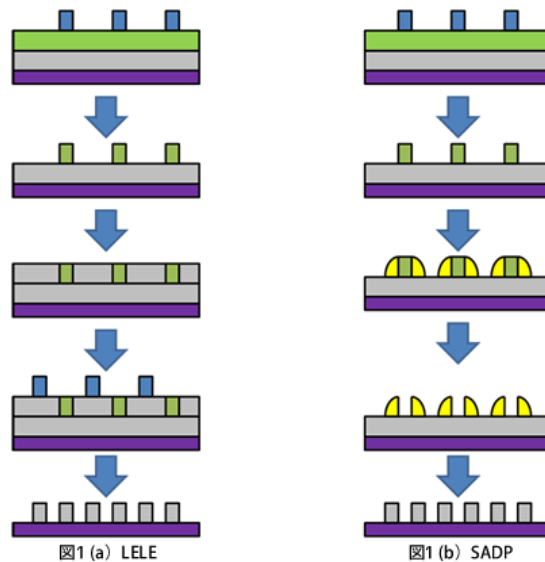


Figure 1.5 LELE and SADP process schematic diagram.



1.4 3D NAND Flash memory HAR etching process technology

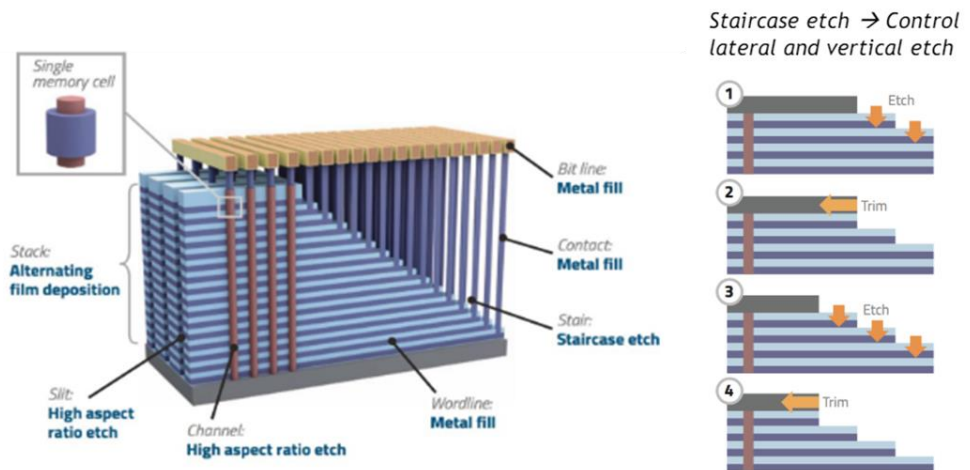
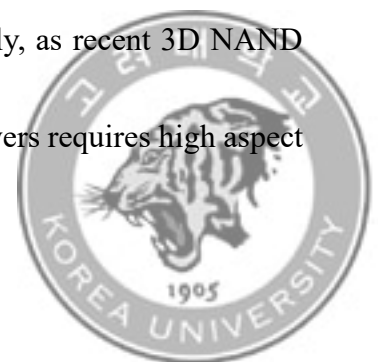


Fig. 1.6 V-NAND flash memory architecture

For example, in the production process of 3D NAND Flash, alternating layers of oxide and nitride thin films are deposited, followed by the deposition of a hard mask and the formation of openings to enable the etching of vertical channels. In this process, it is essential to form holes penetrating the NAND uniformly and consistently from the first layer to the last layer. Particularly, as recent 3D NAND reaches the 2xx layer range, an increase in the number of layers requires high aspect



ratio etching, leading to ongoing research on various methods and equipment development for hole etching. In HARC etching processes, increasing production costs and cycle times are occurring due to long process times. To overcome this situation, methods to increase the selectivity of etching processes are being researched. [12-16] One of these methods is to increase the amount of gas forming polymers to increase passivation. However, excessive polymer formation can lead to etch stoppage. For example, problems encountered during high aspect ratio etching include hole blockage at the top of the pattern, uneven hole sizes at the top of the pattern when etching down to the bottom (necking and bowing), pattern tilting, and pattern closure. [17-26] Due to the challenges of these processes, meticulous control of each process variable is necessary.



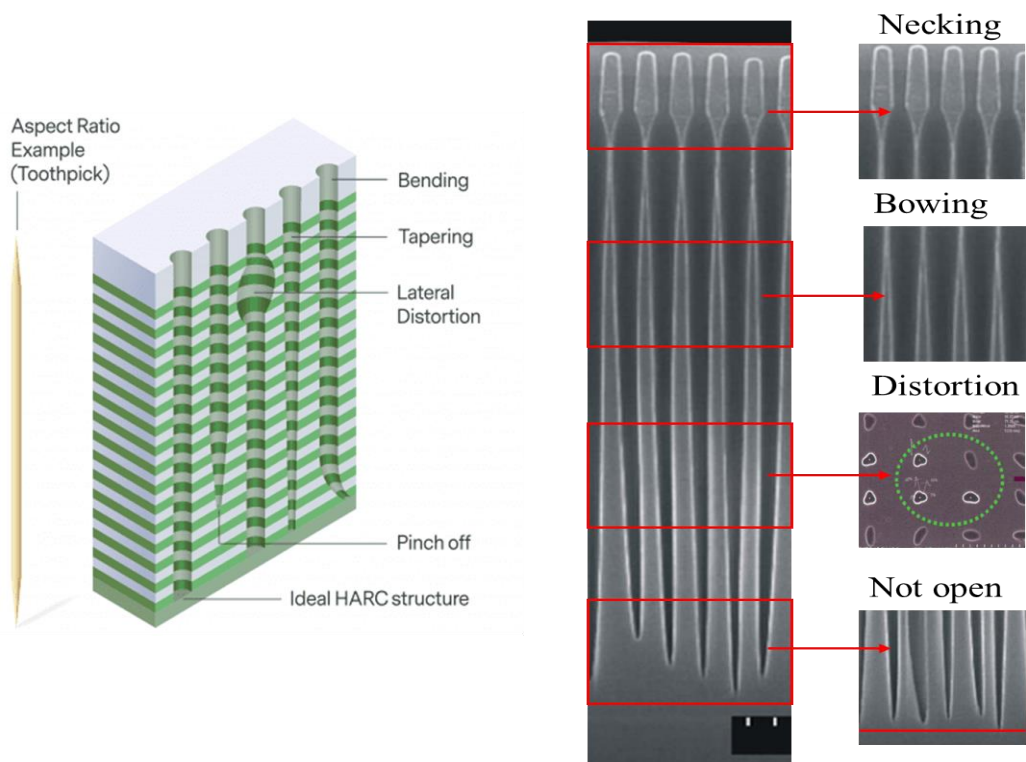


Figure 1.7 Problems arising from the etching profile during high aspect ratio etching.

1.5 Capacitively Coupled Plasma (CCP) and Inductively Coupled Plasma (ICP)

If we were to describe the plasma sources of CCP type and ICP type, CCP type injects neutral gas particles between two anode and cathode electrodes and applies voltage to generate plasma. On the other hand, ICP type generates plasma by applying voltage using a circular coil at the top of the chamber. In CCP type, a high voltage between the lower electrode and the plasma results in high ion energy, allowing operation at high pressures. It also offers flexibility in operating RF frequencies and provides excellent plasma uniformity, enabling large-area processes. However, at low pressures, the gas pressure decreases, causing significant electron losses and resulting in a relatively lower plasma density of around $10^9/\text{cm}^3$ compared to ICP's $10^{11}/\text{cm}^3$. Additionally, the high ion energy in CCP type may lead to substrate damage during processing.



ICP type, however, generates plasma by applying voltage using a circular coil at the top of the chamber. It creates high-density magnetic fields even at low pressures, resulting in a plasma density 10-100 times higher than that of CCP. Moreover, it offers independent control over plasma density and sheath voltage, providing flexibility to adjust individual factors as required for different processes. Additionally, ICP operates at lower pressures, increasing the mean free path and allowing for higher ion bombardment energy compared to CCP. [27,28]

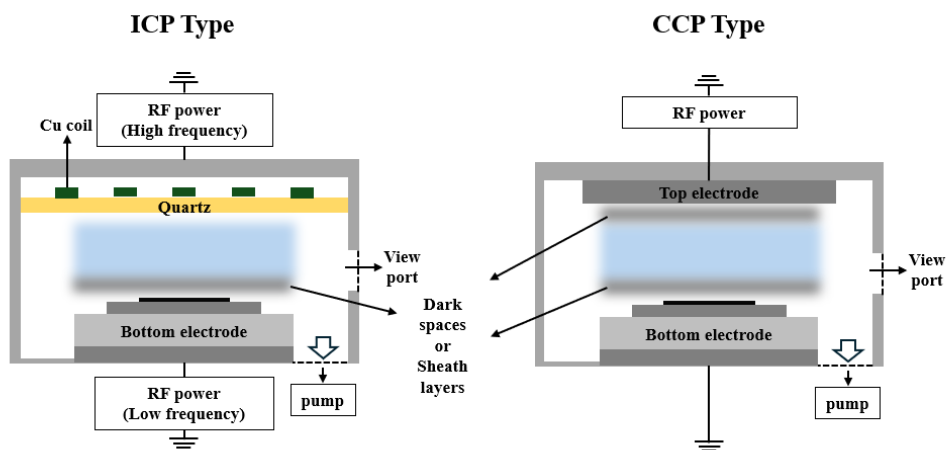


Figure 1.8 Schematic diagram of ICP type and CCP type equipment



1.6 Plasma system used in HAR etching process

The High Aspect Ratio (HAR) etching process has traditionally been performed exclusively using Reactive Ion Etching (RIE) equipment. However, in previous HAR etching processes using Capacitively Coupled Plasma (CCP) or RIE systems, there was a tendency for the etch rate to sharply decrease as the aspect ratio increased. This was due to the inability of ion flux and neutral radicals to reach the bottom of the target material at high aspect ratios [29,30]. Additionally, issues related to profile distortion have been reported as the aspect ratio increases in HAR etching processes [31-34]. Various studies have been conducted to investigate the causes of such profile distortion, which include changes in process gases and pressure, RF power and frequency, and etching profile variations [6,35,36]. The results of these studies suggest a correlation between ion energy, etch rate, and profile distortion due to the adsorbed polymer layer. [17-26]



In today's semiconductor process development, there are numerous studies and challenges related to various profile distortions. However, the most critical condition is to avoid profile distortion caused by ion energy by adjusting the ratio of carbon (C) and fluorine (F) in CF gases for mask and sidewall passivation. Generally, in CF gases, carbon is known to affect the FC polymer, while fluorine determines the effectiveness of the etching. [37] To achieve higher aspect ratio etching, it is necessary to individually control the source power, which contributes to plasma density generation, and the bias power, which affects ion energy. In Inductively Coupled Plasma (ICP), each element can be controlled individually, and it is known to have advantages in etching processes with smaller patterns due to the benefits of the mean free path at low pressure. [38]



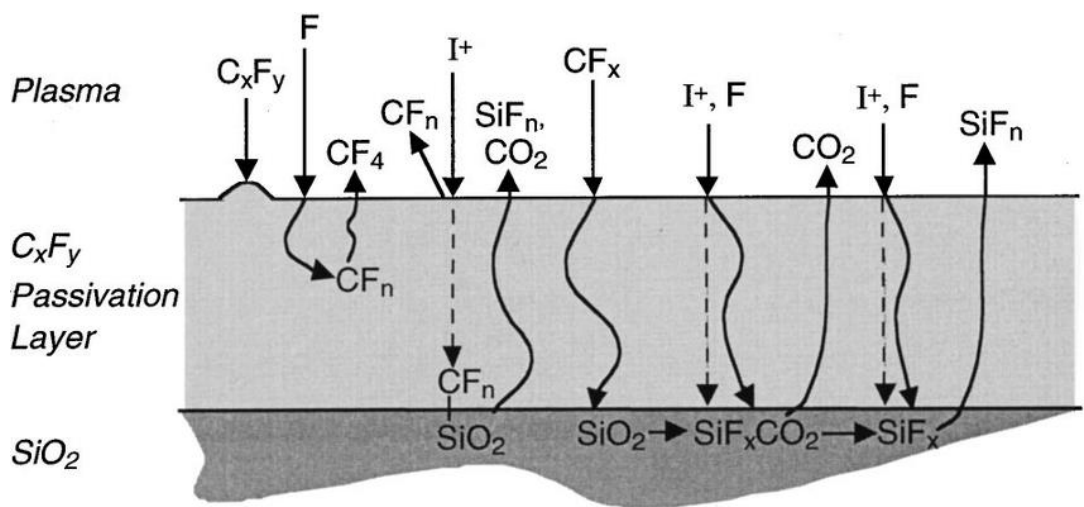
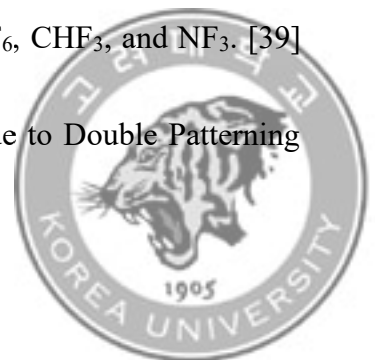


Figure 1.9. Schematic diagram of chemical reaction on the surface during SiO₂ etching

1.7 Perfluorocarbon (PFC)- based gas

Perfluorocarbon (PFC)-based gases are the most widely used gases in semiconductor and display manufacturing processes for etching and cleaning, consisting of compounds of carbon and fluorine. [39] These PFC gases are known to have a high Global Warming Potential (GWP) of over 8000, adversely affecting the ozone layer. [39] Representative PFC gases include CF₄, C₂F₆, C₄F₈, SF₆, CHF₃, and NF₃. [39] Especially, with the increase in patterning process steps due to Double Patterning



Technology (DPT) and Quadruple Patterning Technology (QPT), the usage of PFC gases is expected to continue rising. [40-42] The emission of PFC gases has been steadily increasing over the past decade in tandem with advancements in semiconductor processing technology. [43]

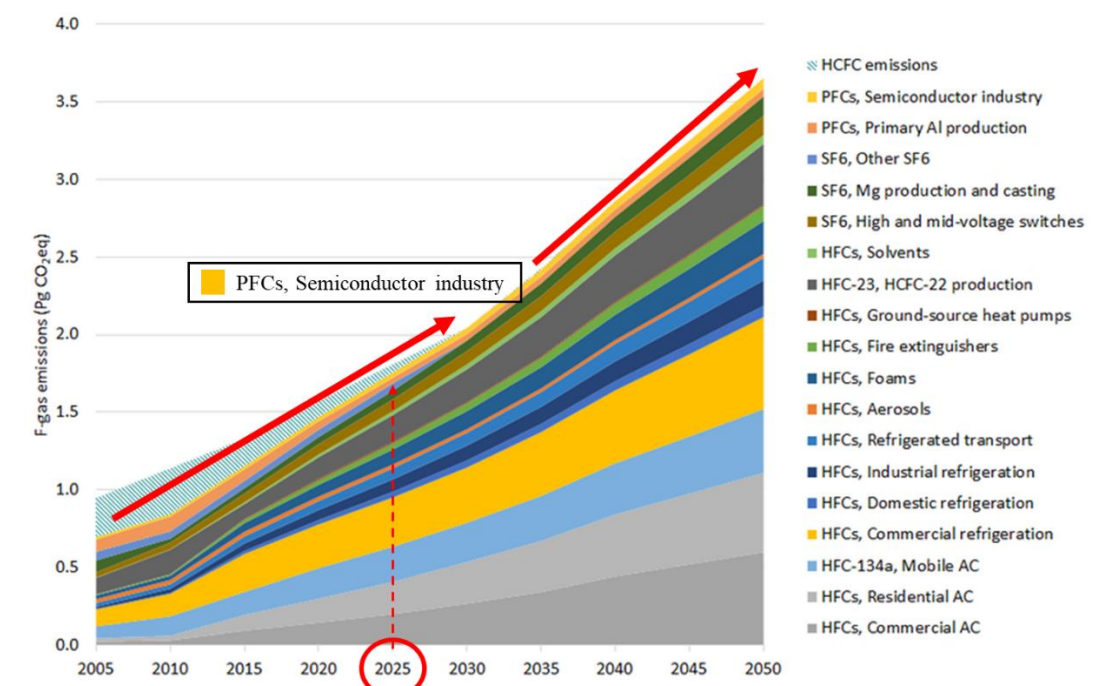


Figure 1.10. PFC gas consumption in the semiconductor industry



1.8 Importance of environmental issues

The Earth's average temperature has generally fluctuated between 1.0°C to -0.5°C.

From the 15th to the 19th century, temperatures were relatively low, but since the 20th century, a distinct upward trend has emerged. Unlike past temperature patterns in the Earth's climate system, modern temperature increases are primarily attributed to greenhouse gas (GHG) emissions. The rapid increase in GHG emissions due to industrialization has accelerated temperature rise. Consequently, the Earth's climate system is experiencing unprecedented and rapid changes, with this global warming phenomenon formally identified in the 1972 report by the Club of Rome. Subsequently, various countries have begun implementing policies and regulations to mitigate climate change by reducing greenhouse gas emissions.



EVOLUTION OF AVERAGE GLOBAL TEMPERATURES

Change in temperatures since 1850, in degrees Celsius

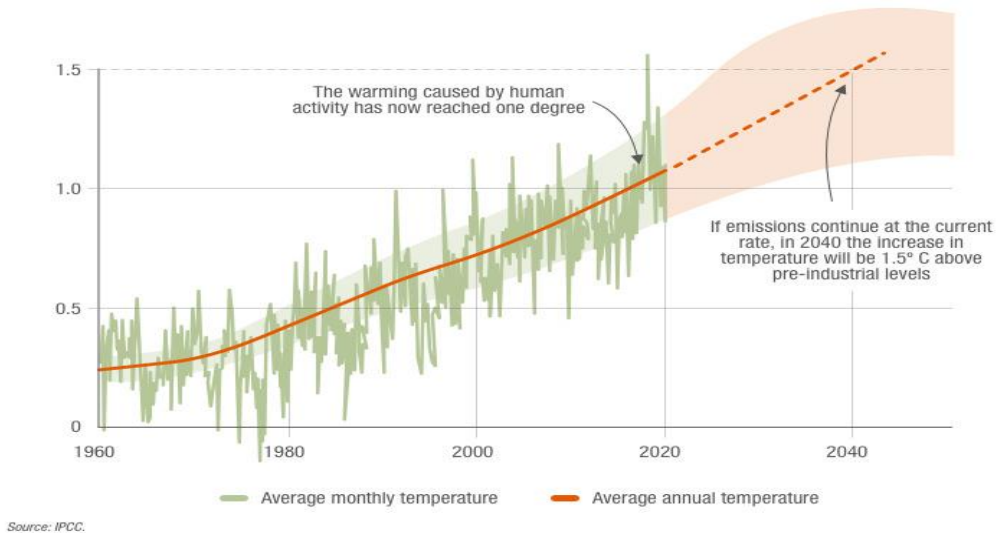


Figure 1.11. Evolution of average global temperatures

The greenhouse gas emissions from the semiconductor industry are expected to increase from the current 11.8 million tons to 14.2 million tons, highlighting the growing need for emission reductions. In response, the World Semiconductor Council (WSC) and the Paris Agreement (2015) have made international efforts through voluntary commitments to reduce perfluorocarbon (PFC) gas emissions. [44,45] The semiconductor industry is striving to minimize GHG emissions by

producing high-efficiency and high-capacity environmentally friendly semiconductors, thereby reducing environmental pollution and resource consumption. [46] A long-term solution to achieve these goals is to replace currently used process chemicals with those that have less environmental impact. [47,48] Various potential solutions that have been recently investigated or are currently under investigation include process optimization, reduction technologies, capture/recovery technologies, and the use of low GWP PFC gas alternatives. [49-52] Particularly, there have been several attempts to replace high GWP PFC gases with low GWP PFC gases and to decompose/recycle them. [51-55] To this end, various gases with low GWP are being tested for application in the semiconductor industry, and efforts are ongoing to reduce the greenhouse effect. Among these, C_4F_6 has emerged as a potential alternative to traditional etching gases due to its low GWP index. [56-58]

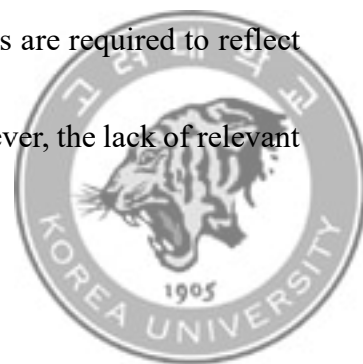


Type of greenhouse gas	Lifetime (year)	GWP
CO ₂	50 – 200	1
CF ₄	50,000	6500
CHF ₃	250 – 390	11700
C ₂ F ₆	10,000	9200
C ₃ F ₈	2,600 – 7,000	7000
C ₄ F ₈	3200	23900
C ₄ F ₆	2day	1

Figure 1.12. The GWP and Lifetime of major gases

1.9 Research objectives

Plasma etching is an essential process step for manufacturing micro and nano electronic devices. High-density plasmas such as helicon, inductively coupled plasma (ICP), and electron cyclotron resonance (ECR) are known to offer many advantages. Given the characteristics of ultra-precision semiconductors, the manufacturing process varies depending on the product specifications. Any changes in the existing process gases and conditions can lead to critical defects and yield loss. Therefore, numerous evaluation procedures and verifications are required to reflect the changed conditions in actual production processes. However, the lack of relevant



research and systematic studies on gas processes and chemical properties poses difficulties in applying them to actual etching processes. Continuous and sustained research and interest in specific fields are essential for the development of etching processes. Moreover, efforts to reduce carbon emissions in the continuously evolving semiconductor industry should be considered alongside technological advancements.

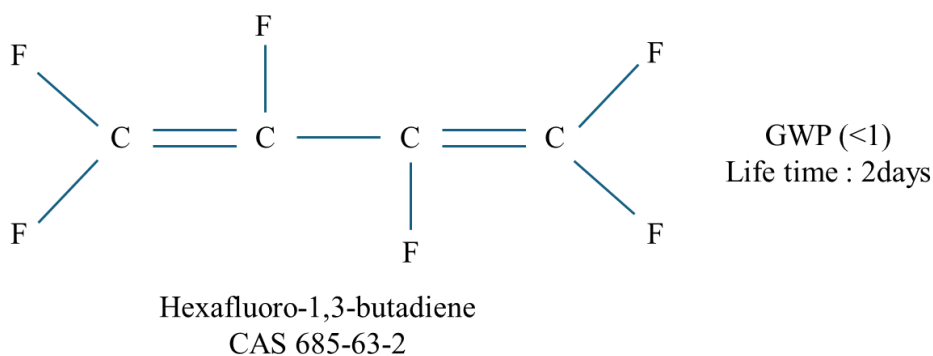
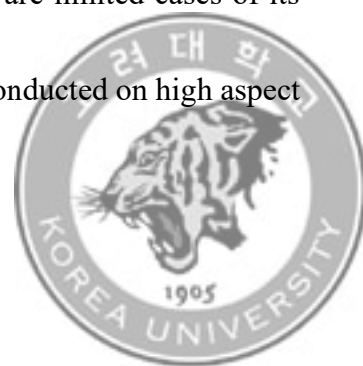


Figure 1.13. Chemical bonding structure of Hexafluoro-1,3-butadiene

C_4F_6 stands out as a promising candidate for overcoming environmental concerns due to its low Global Warming Potential (GWP) index (<1) [56-58]. While this gas is suitable for etching processes due to its low GWP, there are limited cases of its application in etching processes, and no research has been conducted on high aspect



ratio oxide etching using C_4F_6 in ICP systems. To confirm the potential of C_4F_6 as a replacement for conventional etching gases, optimization of plasma etching processes using appropriate operating gases and input process conditions is necessary. Additionally, various experiments and measurements are required to determine the critical process quality factors in semiconductor processes. Therefore, this study investigated the etching mechanism and characteristics of C_4F_6 gas and evaluated its potential application in high aspect ratio etching processes compared to conventional PFC gases.



Chapter 2

Experimental equipment

2.1 Plasma etcher (Inductively Coupled Plasma : ICP)

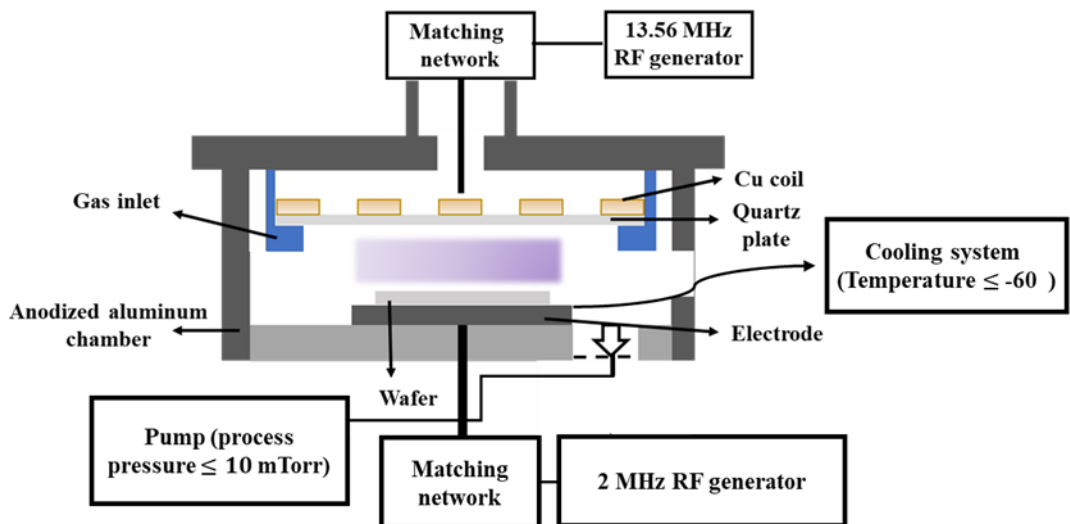
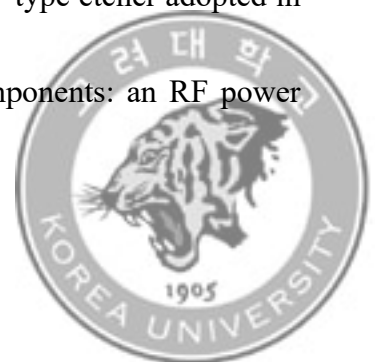


Figure 2.1 ICP etcher schematic diagram 1

Figure 2.1 depicts the schematic diagram of the ICP type etcher adopted in this study. The ICP etcher used consists of four main components: an RF power



supply unit (source or bias), a vacuum chamber, flow and pressure control devices, and vacuum measurement devices. Plasma diagnostic tools such as Langmuir probes and Optical Emission Spectroscopy are installed through the sides of the vacuum chamber and observed through a viewport.

2.1.1 Vacuum chamber



Figure 2.2 anodizing vacuum chamber

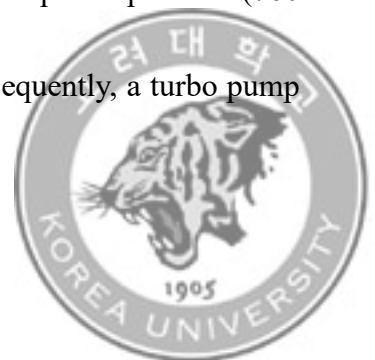
The vacuum chamber is constructed from bipolar oxidized aluminum and is



equipped with ports for plasma diagnostic tools as well as viewport windows located on the sides and front. Additionally, gas lines are provided to transfer process gases into the vacuum chamber. Volatile substances generated during the process are expelled to the outside through an exhaust system consisting of a turbo molecular pump (TMP), dry pump, and scrubber.

2.1.2 Flow rate and Pressure control device

The flow and pressure control system consists of components such as the Mass Flow Controller (MFC), which regulates the flow of gas from the gas cylinder to the vacuum chamber; the Turbo Molecular Pump (TMP) and dry pump, which expel volatile by-products generated during the process; throttle valves to adjust the internal pressure of the chamber; and flow and pressure controllers. During the process, a dry pump is used to decrease the pressure from atmospheric pressure (760 Torr) to 5×10^{-4} Torr, creating a low vacuum condition. Subsequently, a turbo pump



(TMP) is employed to further decrease the pressure to 1×10^{-7} Torr, achieving a high vacuum condition

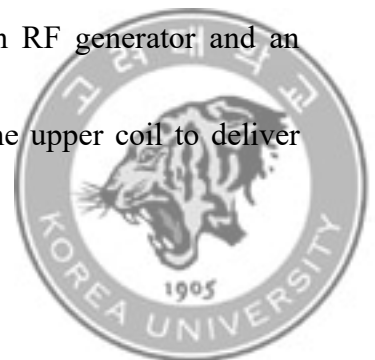
2.1.3 Vacuum measuring device

The vacuum inside the equipment's chamber was measured using Penning, Pirani, and Baratron gauges, and the measured values were displayed on the control panel.

The Pirani gauge and Penning gauge have measurement ranges of atmospheric pressure to 10^{-3} Torr and 10^{-4} Torr to 10^{-8} Torr, respectively. The pressure between the throttle valve and the TMP was measured using a Baratron gauge.

2.1.4 RF supply device

The power supply unit is used to supply RF (13.56MHz, 2MHz) power for plasma generation and maintenance. This device, consisting of an RF generator and an impedance matching circuit, supplies RF power through the upper coil to deliver



energy to the plasma. An impedance matching circuit is installed between the power coil and the RF generator to optimize transmission efficiency. The impedance matching circuit maintains the plasma discharge impedance inside the vacuum chamber close to the output impedance of the RF generator (50Ω). The power supply unit is divided into two components. One is connected to the RF induction coil, and the other is connected to the lower electrode. The induction coil is wound in five turns, inducing a non-contact discharge through inductive coupling using a material with a 10mm thickness of quartz between the coil and the plasma generation area. Additionally, the cooling system for the induction coil effectively controls the heat generated by the coil. When high-frequency power is applied, the cooler is installed independently of the equipment to maintain a constant temperature of the circulating water in the equipment.



2.2 Langmuir probe (Single , Double)

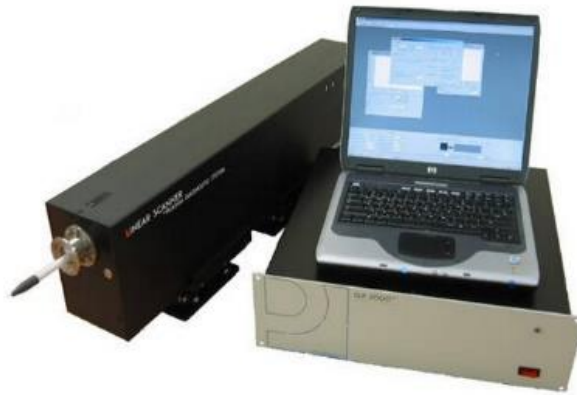
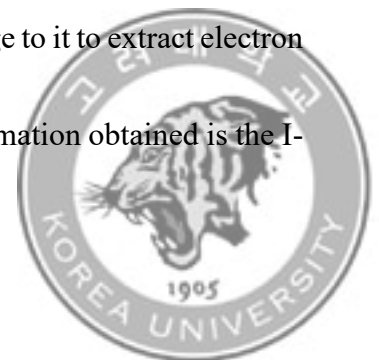


Figure 2.3 Double Langmuir probe

The single Langmuir probe is a device used for measuring plasma parameters. It can measure electron density, ion density, electron temperature, ion saturation current, and electron energy distribution function. This equipment consists of a metal probe, a variable power supply to apply voltage to the probe, and a current measurement tool to measure the current flowing through the probe. The Langmuir probe inserts a metal probe into the plasma and applies voltage to it to extract electron or ion currents from the plasma for analysis. The basic information obtained is the I-



V curve, which represents the relationship between the applied voltage and the current. Analyzing this curve provides information such as ion saturation current density, plasma density, electron temperature, plasma potential, floating potential, and electron energy distribution function. When the probe voltage is increased above the plasma potential (V_p), only electrons enter the probe. The current entering the probe at this point is due to electrons, and in this region, the electron current should remain constant regardless of voltage. However, in reality, the current tends to increase with increasing voltage due to the increase in the effective area of the probe caused by the increase in the sheath area around the probe. When the probe voltage is decreased below V_p , electrons entering the probe must have higher energy than the plasma potential. Electrons with lower energy are reflected by the probe. Therefore, if the electron energy distribution follows a Boltzmann distribution, the electron current in this region appears as an exponential function. Additionally, at certain voltages, the current flowing through the probe becomes zero, which is referred to as the floating potential. If V_p is decreased further below V_f , the electron current



decreases significantly, and the ion current increases. In this region, known as the ion saturation current, the current magnitude does not change significantly even with decreasing voltage. By analyzing the characteristic curve of the Langmuir probe, plasma parameters can be extracted, and the electron energy distribution function can be obtained through calculations, including second derivatives. The single Langmuir probe is a versatile device for measuring various plasma parameters, especially the electron energy distribution function. Due to its advantages, the single Langmuir probe has been actively used in understanding plasma generation mechanisms, and measurement and interpretation methods continue to evolve. Care should be taken in RF plasmas, as the plasma potential oscillating at RF frequencies can cause distortion in the I-V characteristic curves.



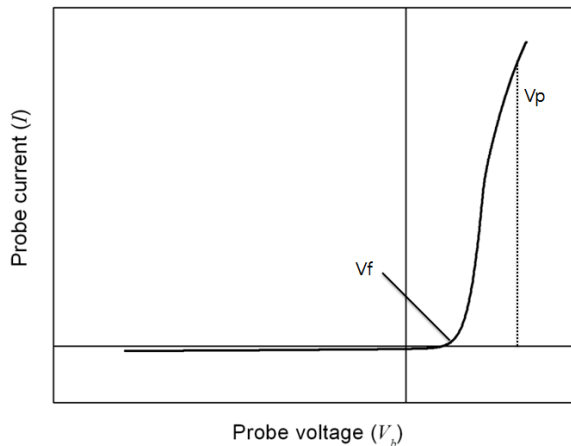


Figure 2.4. I-V characteristic curve of Single Langmuir probe

The double Langmuir probe method is a device that can measure ion density, electron temperature, and ion saturation current. It consists of two metal probes, a variable power supply to apply voltage, and a device to measure the current flowing through the probes. The characteristic curve of the double Langmuir probe also shows the I-V characteristic curve representing the current variation with voltage. The characteristic curve of the double Langmuir probe exhibits an origin-symmetrical shape and has two advantages over the single Langmuir probe. With an electrically isolated structure, unlike the single Langmuir probe, the measurement circuit is



electrically insulated, resulting in less distortion of the I-V characteristic curve due to changes in plasma potential. It can also be used in environments where the plasma potential is not clearly known, such as when the internal surfaces of the chamber are electrically insulated, as in semiconductor and display processes. The current flowing through the double Langmuir probe has the same maximum size as the ion saturation current, so measuring with a small current captures less perturbation to the plasma discharge state. This feature makes it suitable for measurements in process equipment where harsh environments are created by high-power RF power.

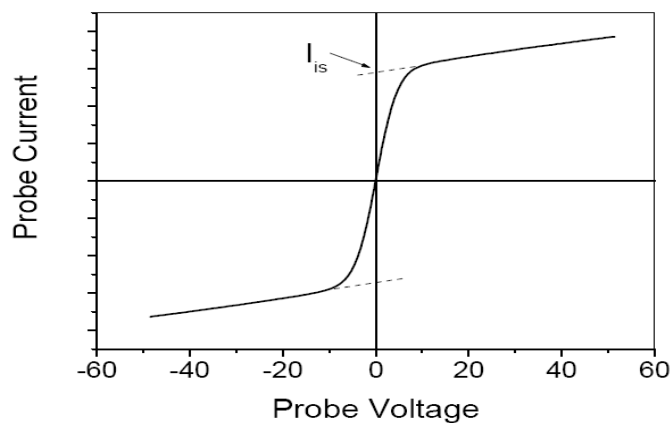


Figure 2.5. I-V characteristic curve of Double Langmuir probe



2.3 Optical Emission Spectroscopy (OES)

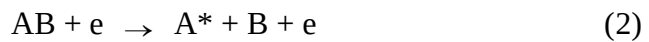


Figure 2.6. Optical emission spectroscopy

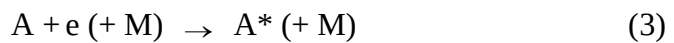
Optical emission spectroscopy is a method used to analyze elements by measuring the specific wavelengths of light emitted due to changes in the electronic states of the elements. This method has the advantage of being able to measure without passing through the chamber, thus not affecting the plasma generated inside the chamber. Through the analysis of specific wavelengths, it identifies the elements present in the plasma and estimates the quantity of elements by varying the intensity of light. However, this method has limitations in that it cannot measure plasma species that do not emit light. Additionally, since the emission intensity of each



plasma species is not directly proportional to its density, it is difficult to directly compare the densities of different plasma species.



Intense light is emitted from excited species through ion collision processes.



As the excited species A^* transitions back to its ground state (A), light corresponding to the energy difference is emitted.



Optical emission spectroscopy is generated by electron collision and dissociation processes in plasma. Light is emitted from excited states during ion detachment



processes. Here, the excited state A^* represents the emitting species, while $e(+M)$ denotes neutral species, negative ions, electrons, third bodies, or surfaces in contact with the plasma. To perform optical spectroscopy, appropriate optical systems and detectors are required to measure the emitted light from the plasma. Light generated in the plasma is transmitted to a spectrometer, passes through a diffraction grating, and is spectrally measured. At the exit of the spectrometer, only light of specific wavelengths is transmitted to the detector, generating a photocurrent proportional to its intensity. For plasma species emitting at specific wavelengths, identification can be achieved through the wavelength of the emitted light. However, since each plasma exhibits a variety of spectral lines, identifying a specific plasma can be challenging. For this reason, optical spectroscopy provides useful information mainly on the light emitted by single or binary particles, as their spectra are simpler and narrower compared to multi-atomic species. Additionally, due to variations in light intensity depending on the plasma state, quantitative interpretation can be challenging.



2.4 X-ray Photoelectron Spectroscopy (XPS)

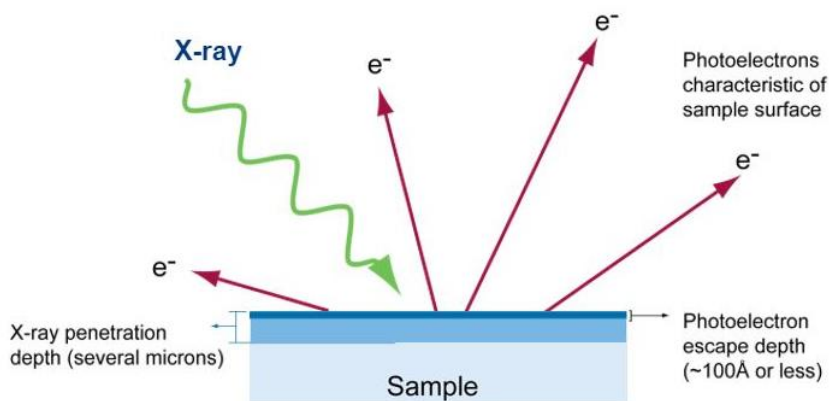
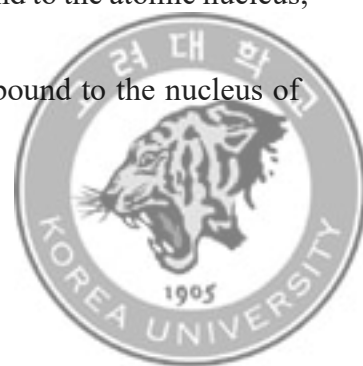


Figure 2.7 X-Ray photoelectron analysis schematic diagram

X-ray photoelectron spectroscopy (XPS) is particularly useful for acquiring information about very thin surface layers within 100 angstroms. It is often known as Electron Spectroscopy for Chemical Analysis (ESCA). When X-rays (photons) with specific energy interact with the sample surface, photoelectrons are emitted from the surface atoms. By measuring the kinetic energy of these photons, we can determine the binding energy (BE) of the inner electrons bound to the atomic nucleus, which is the energy required to emit the internal electrons bound to the nucleus of



the atom. This allows us to identify the surface components and understand their chemical bonding states. The binding energy, characteristic of the atoms emitting the photoelectrons, plays a crucial role in identifying the components. Additionally, qualitative analysis of the sample can be performed through binding energy, and the bonding states of atoms can be investigated through changes in binding energy. The binding energy of photoelectrons is useful for identifying elements and evaluating the properties of samples.



2.5 Scanning Electron Microscope (SEM)

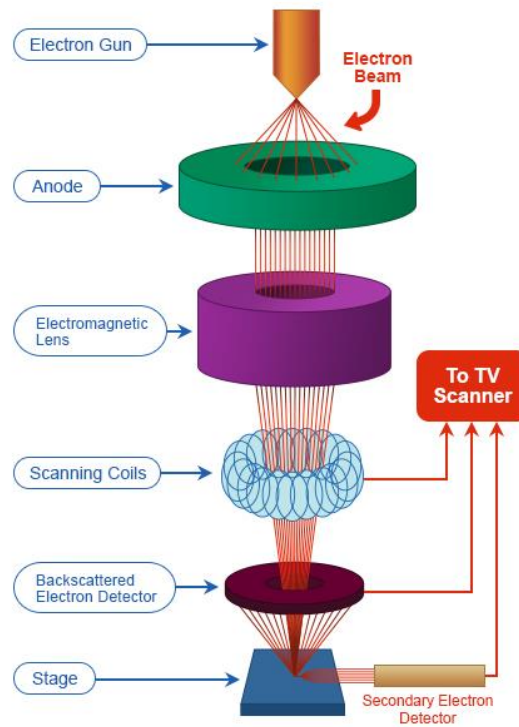


Figure 2.8 Overview of Scanning Electron Microscope (SEM) Analysis schematic

diagram



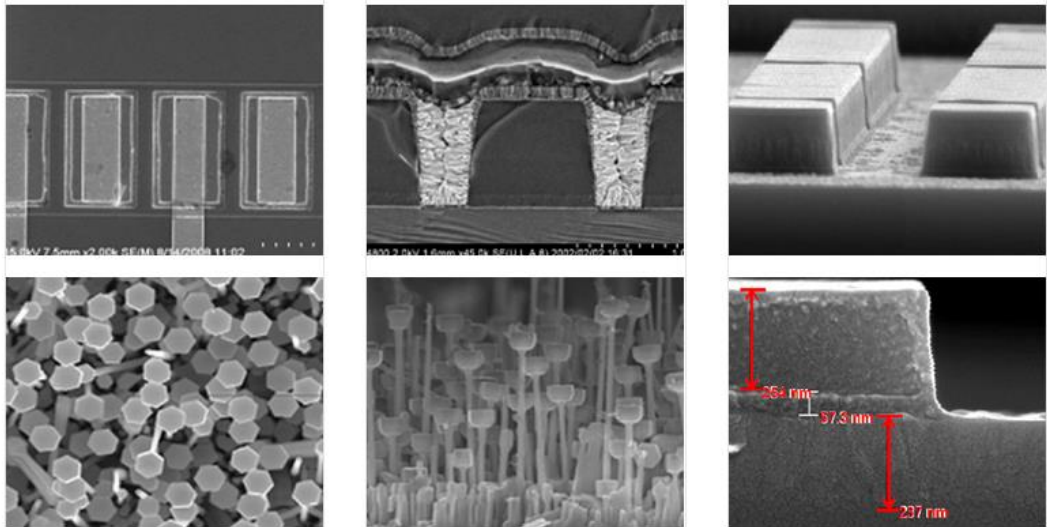
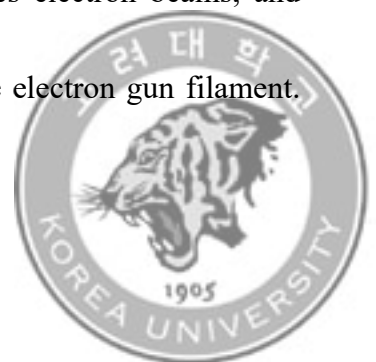


Figure 2.9 Function of Scanning Electron Microscope (SEM)

Scanning Electron Microscopy (SEM) is primarily used to observe the fine structure and morphology of solid samples, thanks to its capability to easily visualize features in three-dimensional (3-D) images. It allows for the high-resolution observation of complex surface structures and crystal forms in three dimensions. The instrument consists of a condenser lens, an objective lens, an electron gun that generates electron beams, a focusing lens that concentrates electron beams, and injection coils that control the path of electrons exiting the electron gun filament.



The electron beam is focused on the surface of the conductive sample to form a concentrated electron beam area, which is then observed using electromagnetic lenses. This method forms images through scanning and observation. Various secondary emissions occur during this process, but the image is displayed to the observer using secondary electrons generated from the interaction between the sample and the electron beam. At this point, the brightness of the image is proportional to the number of secondary electrons generated by the interaction between the electron beam and the sample. Since the sample is monitored through electrons, distortion may occur due to collisions of electrons if air is present internally. Therefore, observations should be made under vacuum conditions (10^{-5} Torr) to prevent this.



3.1 Etching process

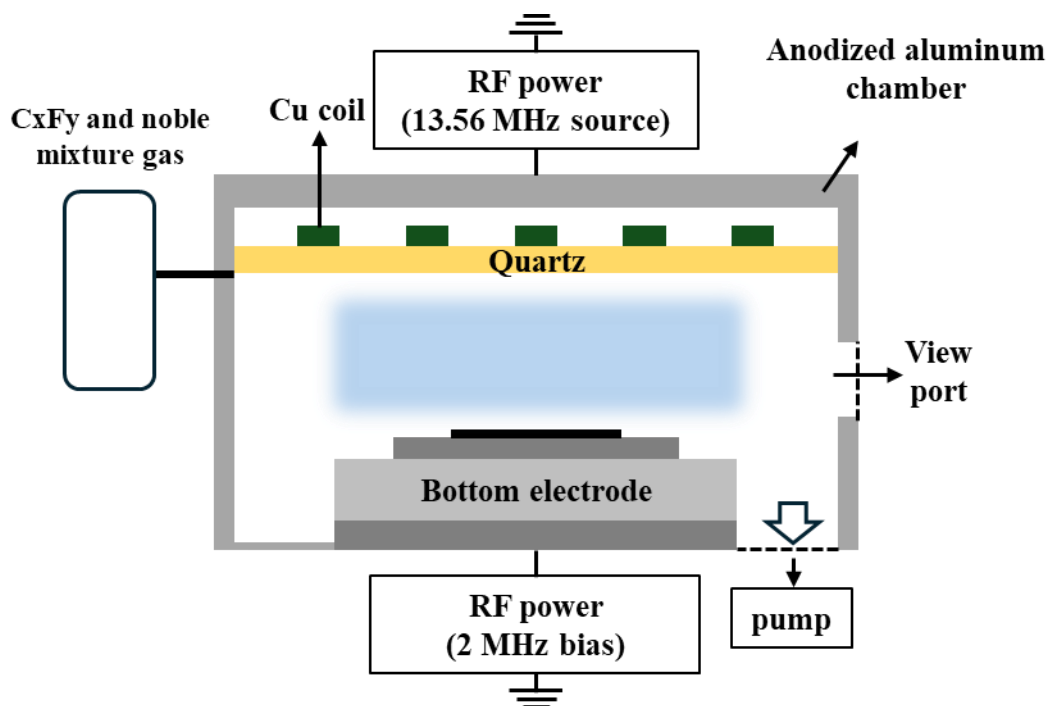
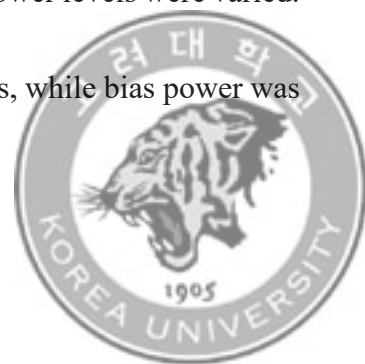


Figure 3.1 ICP etcher Schematic diagram 2

Prior to this study, etching experiments were conducted in a planar ICP reactor as described in previous research [59-61]. A 13.56 MHz RF generator (AE, Paramount 1513) and a 2 MHz RF generator (Advanced Energy, CESAR RF Power Generator) were connected to the lower electrode to establish $-V_{dc}$. The IPA (isopropyl alcohol) flow cooling system of the lower electrode was used to maintain the temperature at approximately 17°C . The experiments were performed varying the $\text{CF}_4/\text{C}_4\text{F}_6$ flow rates, source power, and bias power. Additionally, a constant total gas flow rate of 140 sccm ($\text{CF}_4 + \text{C}_4\text{F}_6 + \text{He}$) was maintained, with a gas pressure set at 10 mTorr. In the flow rate variation experiments, the $\text{C}_4\text{F}_6/(\text{CF}_4 + \text{C}_4\text{F}_6)$ ratio was varied across 5 parameters ranging from approximately 37.5% to 67.5% (total $\text{CF}_4 + \text{C}_4\text{F}_6 = 40$ sccm), while He was fixed at 100 sccm. The source power and bias power were fixed at 150W and 400W, respectively. For the variations in source power and bias power, the $\text{C}_4\text{F}_6/(\text{CF}_4 + \text{C}_4\text{F}_6)$ ratio was fixed at 56.25% (total $\text{CF}_4 + \text{C}_4\text{F}_6 = 40$ sccm), with He fixed at 100 SCCM, and only the power levels were varied. Source power was varied from 50W to 200W in 4 parameters, while bias power was



varied from 300W to 600W in 4 parameters for comparison.

3.2 Sample preparation

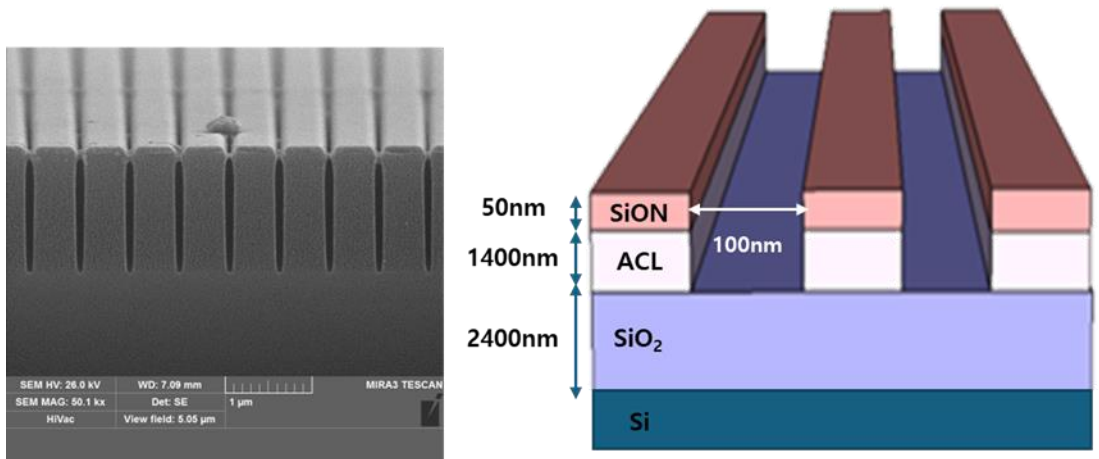
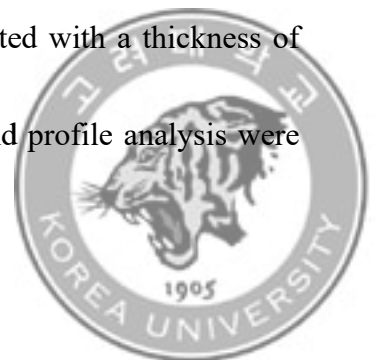


Figure 3.2 Reference sample SEM profile and schematic

The sample consists of a Si-based 300mm wafer with a 2400nm SiO₂ layer deposited on it, followed by a 1400nm thick Amorphous Carbon Layer (ACL) Mask. On top of this, SiON for mask pattern formation is deposited with a thickness of 50nm. The pitch width is 500nm, and etching processes and profile analysis were



conducted on line patterns with a width of 100nm. For the etching experiments, coupon samples measuring 2x2 cm were used at the center of the lower electrode. The use of small sample sizes allowed us to ignore loading effects and provided an etching system controlled by heterogeneous process dynamics. Prior to the etching process, it was confirmed that there were no significant differences in the obtained I-V curves and relevant plasma parameter measurements depending on the presence of the sample loading. Therefore, regardless of the presence of the sample, the sensitivity of plasma parameters during the etching process can be disregarded, and the gas phase can be considered as a permanent source of active species governed by heterogeneous process dynamics.

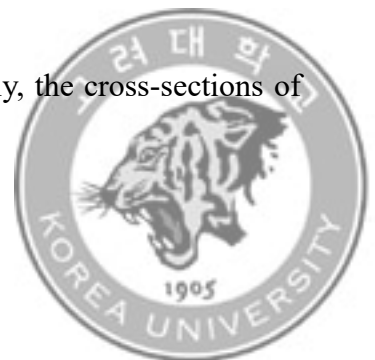


3.3 Analysis of etch properties



Figure 3.3 Scanning Electron Microscope(MIRA3 LMH, TESCAN)

In this study, to investigate the etching profile of SiO_2 , a reference sample was prepared by depositing SiO_2 followed by ACL, with thicknesses of 2400nm and 1400nm, respectively, on a Si substrate as described earlier. The sample consisted of line patterns with a width of 100nm and intervals of 500nm. The hard mask-assisted reactive (HAR) etching process was conducted. Additionally, the cross-sections of



the profiles were measured using a Field-emission scanning electron microscope (MIRA3 LMH, TESCAN).

3.4 Plasma diagnosis

The gas composition for plasma parameters was measured using Double Langmuir probe diagnostics and Optical emission spectroscopy. The Double Langmuir probe (DLP2000, Plasmart Inc., Daejeon, Korea) provided information on electron temperature (T_e) and ion current density (J_{is}). The processing of the raw I-V curves was based on well-established procedures described in the referenced papers [46-47]. To minimize experimental errors caused by the deposition of fluorocarbon polymers on the probe tips, Ar + O₂ plasma (in a 1:1 ratio) was used for cleaning before and after measurements. Previous studies have demonstrated the effectiveness of these procedures in obtaining accurate diagnostic results. [62-64]





Figure 3.4 Optical Emission Spectroscopy (Avaspec-3648, Jinyoung Tech, Korea)

The intensity of light emitted from active species in the plasma was measured using Optical Emission Spectroscopy (OES). For this purpose, a monochromator (Avaspec-3648, Jinyoung Tech, Korea) covering the wavelength range of 200 to 800 nm was employed. Specific wavelengths of CF (255.253 nm), CF₂ (340.129 nm), CF₃ (610.144 nm), F (703.067 nm), and He (501 nm) were utilized [59,60,65]

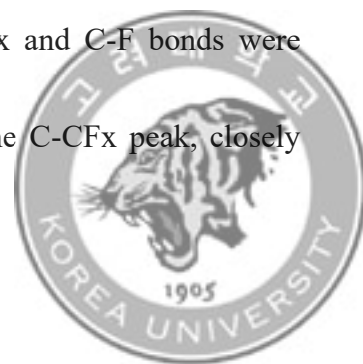


3.4 Surface analysis



Figure 3.5 X-Ray photoelectron spectroscopy(K-Alpha, Thermo VG, UK)

The chemical composition of the SiO_2 and ACL surfaces after the plasma process was examined using X-ray Photoelectron Spectroscopy (XPS) with Mg Ka 1253.6 eV radiation, operated at 260W, employing K-Alpha instrument (Thermo VG, UK). The binding energy was calibrated using the C(1s) peak at 285.4 eV. Furthermore, Deconvolution of the narrow spectrum of C(1s) allowed the extraction of three peaks at binding energies corresponding to C-C, C-CFx, and C-F bonds at 284.7, 286.7, and 289.1 eV, respectively. Polymers composed of C-CFx and C-F bonds were predominantly observed in the oxide film. Additionally, the C-CFx peak, closely



related to polymer thickness, showed a strong correlation with the etching rate of the oxide film. [66-68]



4.1 Effect of gas flow ratios

4.1.1 Etching Characteristics

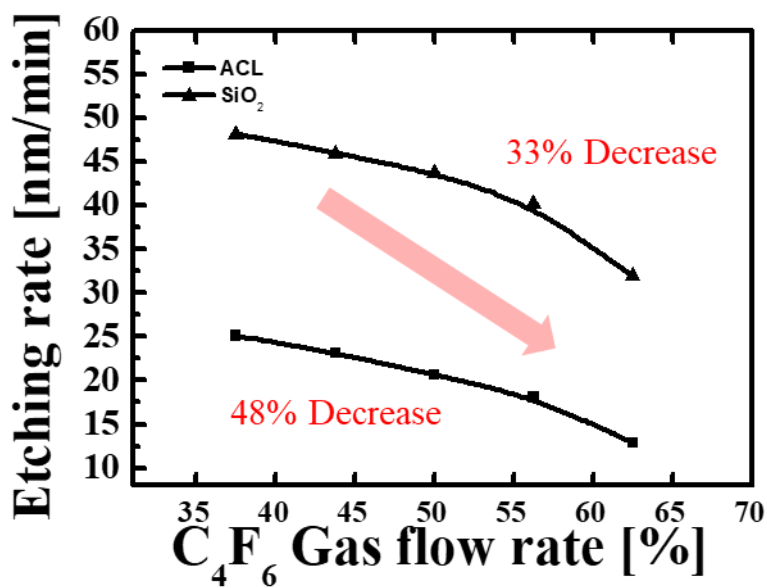
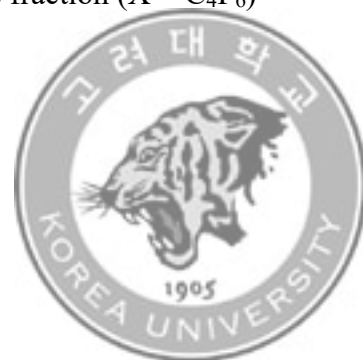


Figure 4.1 SiO_2 , ACL etching rate according to the X gas fraction (X = C_4F_6)



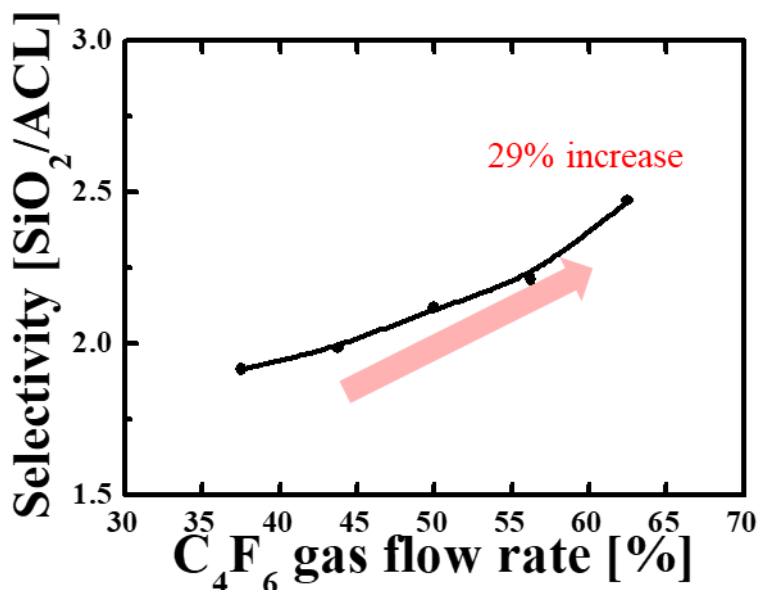
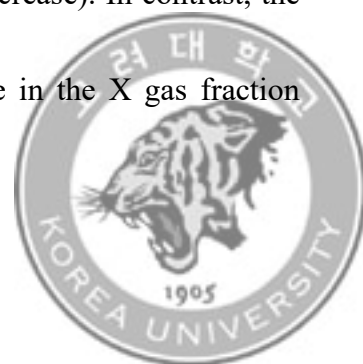


Figure 4.2 Selectivity (SiO_2/ACL) according to the X gas fraction ($X = C_4F_6$)

Figure 4.1 illustrates the influence of the X gas fraction in $CF_4 + X + He$ ($X = C_4F_6$) gas mixture on the etching rate of SiO_2 films. As the X gas fraction increases in the mixed plasma, the etching rate of SiO_2 films decreases (e.g., for 37.5%-62.5% SiO_2 films, the etching rate decreased from 48.155 to 31.922 nm/min, a 33% decrease; for ACL, it decreased from 25.12 to 12.902 nm/min, a 48% decrease). In contrast, the etching selectivity (SiO_2/ACL) increased with the increase in the X gas fraction



(from 1.91 to 2.47 for 37.5%~62.5%, a 29% increase). The etching of Si and Si-based films is determined by the predominant active species, the volatility of reaction products, and the thickness of deposited FC polymers. [63,69] The decrease in the etching rate of SiO₂ films could be attributed to two main factors. Firstly, it may be due to the corresponding changes in the density of reactive neutral species resulting from variations in plasma parameters. Secondly, it may be attributed to changes in FC polymer deposition associated with the F/C ratio within the mixture. [45]



4.1.2 Plasma diagnostic

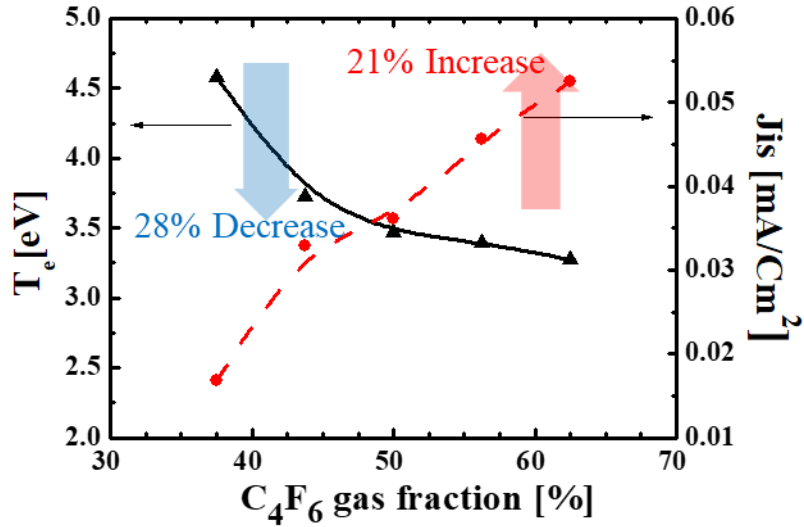
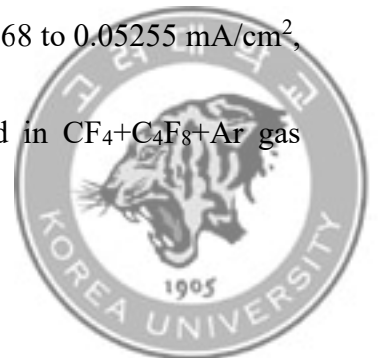


Figure 4.3 T_e & J_{is} according to X gas fraction (X= C_4F_6)

Figure 4.3 illustrates the values of electron temperature (T_e) and ion current density (J_{is}) as the flow rate of X gas (X = C_4F_6) increases. With the increase in the flow rate of X gas (X = C_4F_6), the electron temperature (T_e) decreases, while the ion current density (J_{is}) increases (e.g., for the increase from 37.5% to 62.5%, T_e decreased from 4.57 to 3.27 eV, a 28% decrease, and J_{is} increased from 0.0168 to 0.05255 mA/cm², a 21% increase). Similar observations have been reported in $CF_4+C_4F_8+Ar$ gas



mixtures after replacing C_4F_8 with CF_4 [63,69]. Therefore, the increase in X gas may be related to the increase in ionization rate coefficients (as the number of particles with low ionization thresholds increases). As CF_4 transitions to C_4F_6 -rich plasma, CF_4 and F change to CF_2 and CF_1 , respectively [63,69], leading to more ionization with the increase in C_4F_8 flow rate, resulting in electron temperature loss [70].

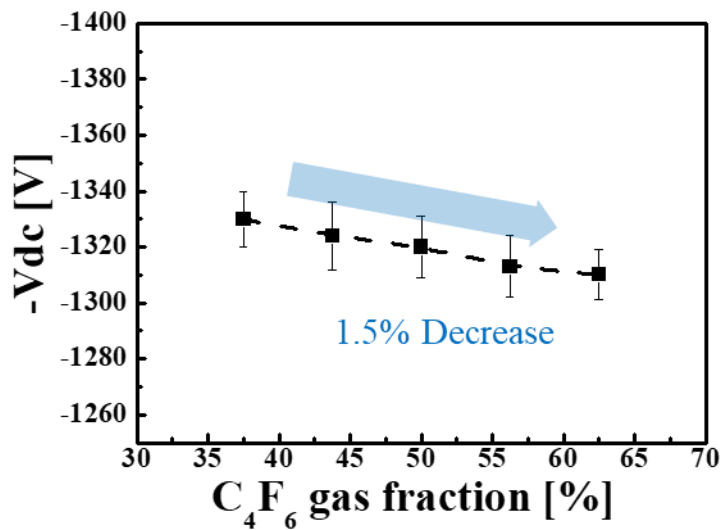


Figure 4.4 $-V_{dc}$ according to X gas fraction (X= C_4F_6)



Figure 4.4 illustrates the variation in $-V_{dc}$ as the X gas fraction increases. There is a decreasing trend as the X gas fraction increases (a decrease of 1.5%). However, the difference in values is minimal, and it appears that the value of V_{dc} remains almost unchanged. The value of $-V_{dc}$ is associated with plasma density. This increase in plasma density is due to the higher ionization resulting from the increased fraction of gas with lower ionization thresholds [71,72]. As plasma density increases, the plasma potential decreases, leading to a narrowing of the sheath region. Consequently, the region where electrons are accelerated decreases, resulting in a decrease in the value of $-V_{dc}$ [73].



4.1.3 OES Characteristics

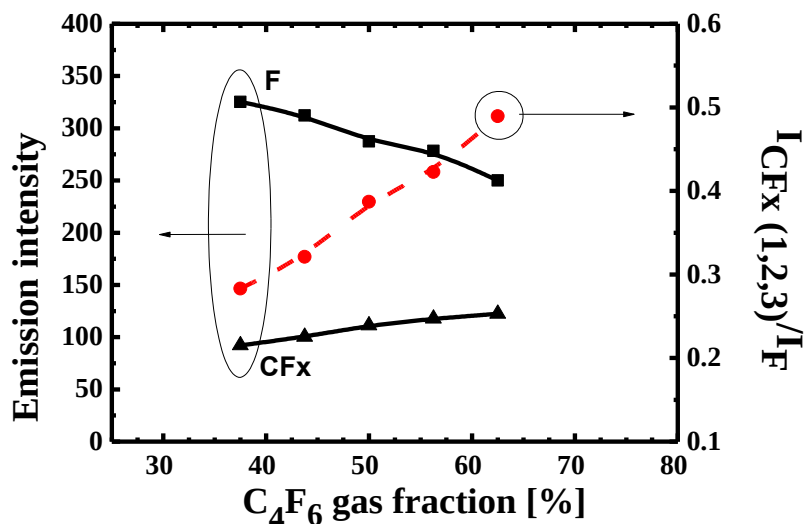
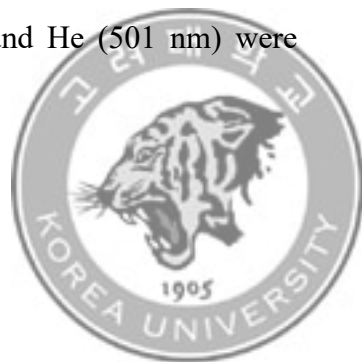


Figure 4.5 Emission intensity according to the X gas fraction (X=C₄F₆)

Figures 4.13 and 4.14 present the optical emission intensity data obtained through Optical Emission Spectroscopy (OES) to investigate the behavior of chemical reactive species such as polymer and fluorine in a CF₄+C₄F₆+He mixed plasma as the X gas flow rate increases (X=C₄F₆). Specific wavelengths of CF₁ (255.253 nm), CF₂ (340.129 nm), CF₃ (610.144 nm), F (703.067 nm), and He (501 nm) were utilized. [60,65,74]



As a result, an increase in CF_x species ($x=1,2$), contributing to polymer formation, was observed with the increase in C_4F_6 gas. Additionally, a decrease in F intensity was noted. With a higher proportion of C_4F_6 gas conditions, it provides a higher radical density of CF_x ($x=1,2$), corresponding to a higher polymerization flux, thus leading to a higher deposition rate of fluorocarbon polymer films. Fluorocarbon polymers are known to increase surface passivation, thereby inhibiting etching [67]. These results indicate that the decrease in the etch rate of SiO_2 is attributed to the increased deposition of FC polymer layers on the SiO_2 film surface and the reduction in F atom dissociation throughout the plasma etching process.



4.1.4 Etched profile

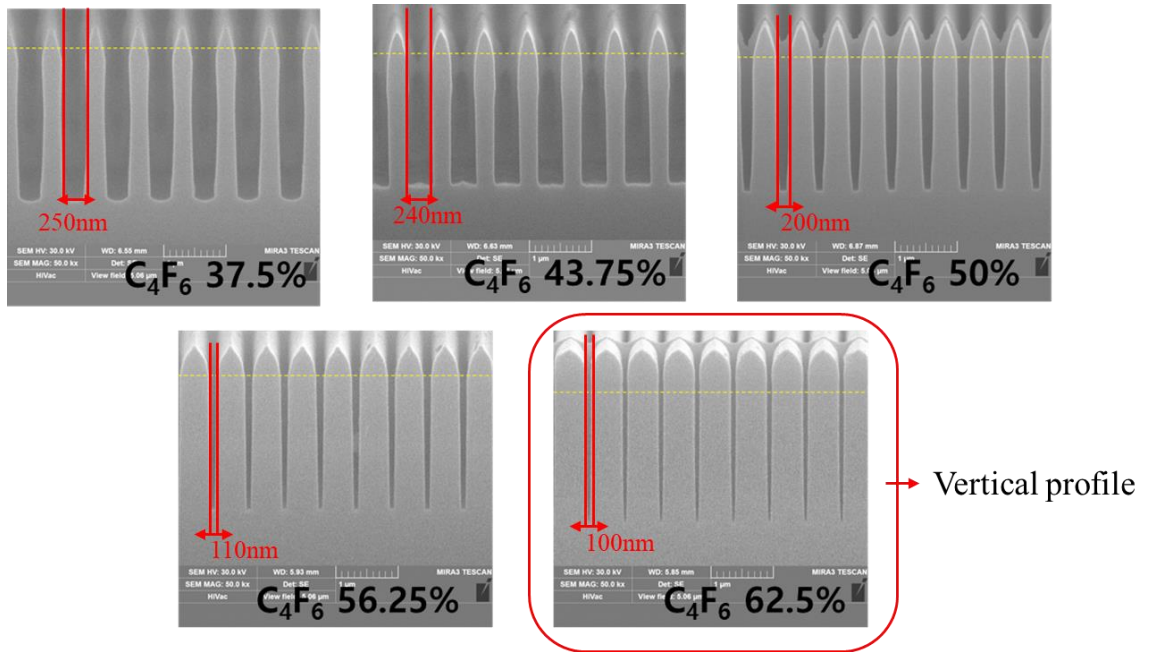


Figure 4.6 Etched profile in the X gas fraction ($X=C_4F_6$)

Figure 4.6 depicts the etch profile images as a function of the X gas flow rate ($X = C_4F_6$) in $CF_4 + C_4F_6 + He$ composite plasma. Post-plasma etching FE-SEM images of a 100 nm line pattern with a width of 500 nm are presented. Plasma etching experiments were conducted using the same etching conditions for $CF_4 + C_4F_6 + He$ mixed plasma, except for adjusting the etching time to achieve the same etch depth



at different X gas flow rates. The SiO₂ film etch depth ranged from 2100 to 2200 nm, and the most vertical profile was obtained at the C₄F₆ 62.5% condition. It was observed that the facet phenomenon of the mask decreased with an increase in the C₄F₆-rich condition. Additionally, with an increase in the C₄F₆-rich condition, the widening of the linewidth observed in the CF₄-rich condition gradually decreased (e.g., the 100 nm width pattern widened up to 250 nm at the CF₄ 62.5% condition). This result may be attributed to the formation of polymer on the etched sidewalls. [75] C₄F₆-containing plasma is characterized by a higher probability of polymer generation compared to CF₄ plasma (as evident from the data above) and the formation of sidewall passivation that reflects high-energy ions. These characteristics not only prevent an increase in linewidth due to sidewall passivation but also prevent mask erosion by forming passivation on both the ACL mask surface and sidewalls. [76,77]. Thus, utilizing C₄F₆ instead of the widely used CF₄ for high aspect ratio oxide etching in each etching process seems feasible for achieving the desired etching goals.



4.1.5 Surface Analysis

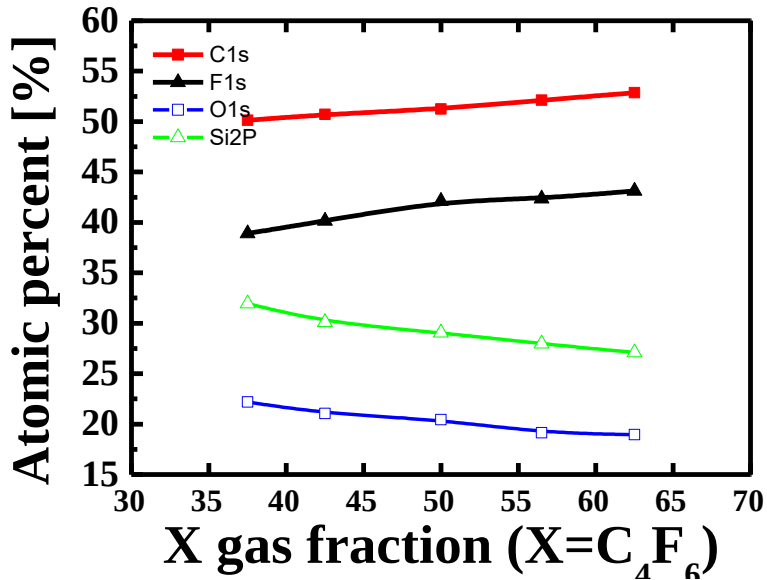


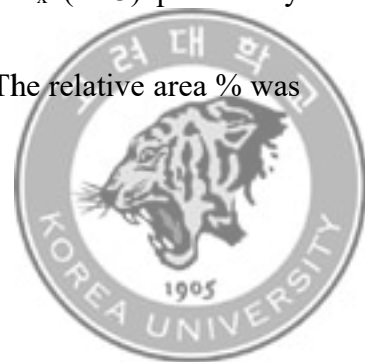
Figure 4.7 XPS peak spectra of the SiO₂ surface in the CF₄+C₄F₆+He plasma when the X gas fraction change

CF ₄ /C ₄ F ₆ /He AT %	C1s	F1s	O1s	Si2P
25/15/100	50.12	38.9	22.19	31.96
23/17/100	50.71	40.15	21.05	30.11
20/20/100	51.25	42.123	20.45	29.05
17/23/100	52.11	42.358	19.14	27.98
15/25/100	52.87	43.125	18.95	27.11

Table 1. Atomic % of XPS etched SiO₂ surface at fraction of C₄F₆.



To investigate how the atomic compositions of SiO₂ and ACL surfaces changed as the X gas flow ratios increased ($X = \text{C}_4\text{F}_6$), XPS analysis was conducted in CF₄ + C₄F₆ + He mixed plasma. Figure 4.19 represents the atomic % of each peak C1s, F1s, O1s, Si2p as a function of the X gas ratio. As the X gas ratio increased, the atomic ratios of C1s and F1s on the SiO₂ surface gradually increased, while the atomic ratio of Si2p decreased. It is well-known that C1s and F1s are associated with residual fluorocarbon etching residues on the surface. To further investigate these peaks, Deconvolution of the spectrum was performed. Figure 4.20 illustrates the Deconvolution results of the peaks obtained from C₄F₆. Deconvolution was performed to align with the binding energies of all XPS data. Bond energies of C-Si, C-C, C-CF_x, C-F, C-F₂, C-F₃ were discovered, which were analyzed only for C 1s (to confirm the presence of FC etching residues on the surface). The observed peaks can be assigned to C-Si (283.2eV), C-CF_x (287.0eV), C-F₁ (289.3eV), C-F₂ (291.4eV), C-F₃ (293.3eV) [76,80-82]. The C-CF_x and C-F_x ($X < 3$) peaks may originate from polymers formed on the surface [76,80-82]. The relative area % was



calculated by multiplying the ratio of each bonding state corresponding to the C 1s ratio and then dividing the sum by 100.

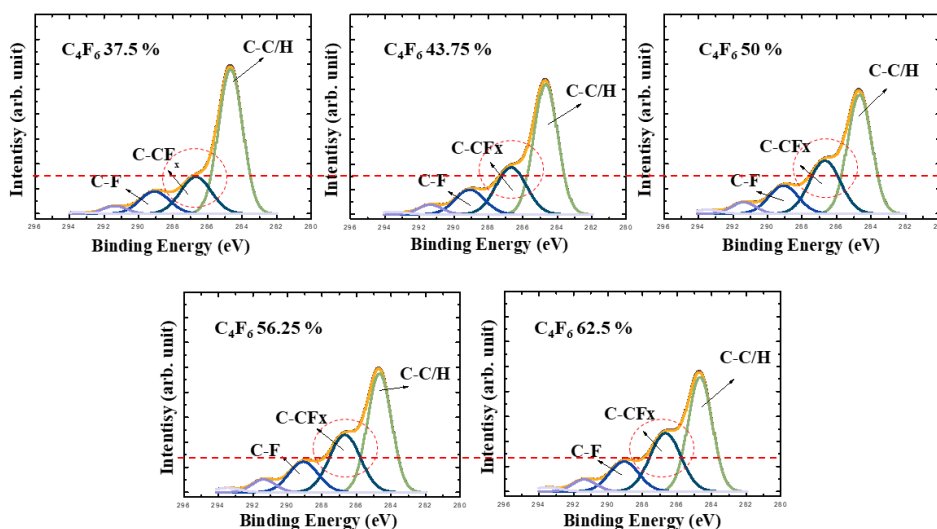
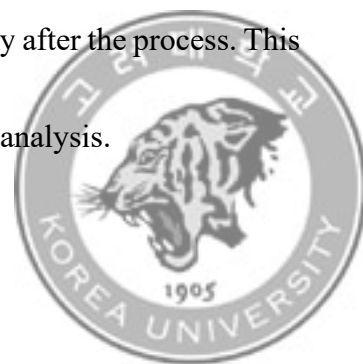


Figure 4.8 C1s deconvolution in the X gas fraction change (X=C₄F₆)

As the C₄F₆ gas ratio increased, the C-C bond gradually decreased, while the C-CF_x and C-F bonds increased. The decrease in the C-C bond ratio with increasing C₄F₆ gas ratio indicates its susceptibility to atmospheric contamination, as evidenced by its persistence even without breaking the vacuum immediately after the process. This suggests that the C-C peak can serve as a reference for XPS analysis.



Furthermore, the C-CF_x and C-F bonds noticeably increased as the X gas ratio increased. This indicates that with higher C₄F₆ abundance in the plasma, there is an increase in polymer formation, as evidenced by the increase in the C-CF_x peak.



4.2 Effect of Source/Bias power

4.2.1 Etching characteristics

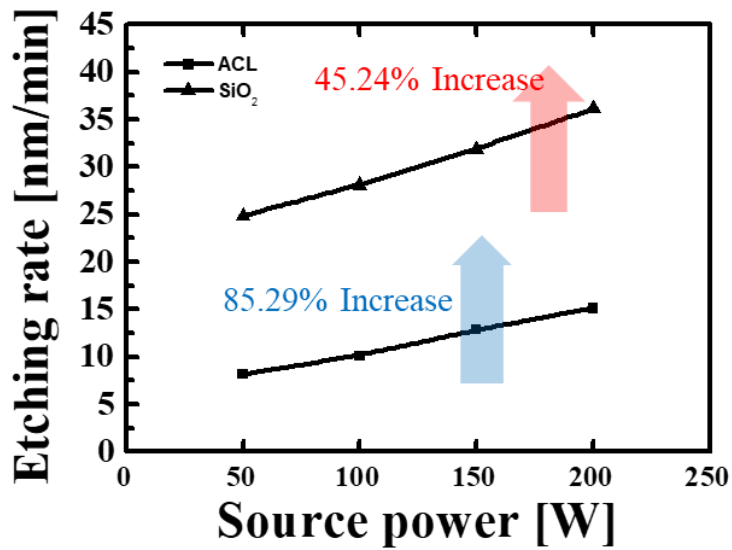


Figure 4.9 SiO₂, ACL etching rate according to the Source power increase

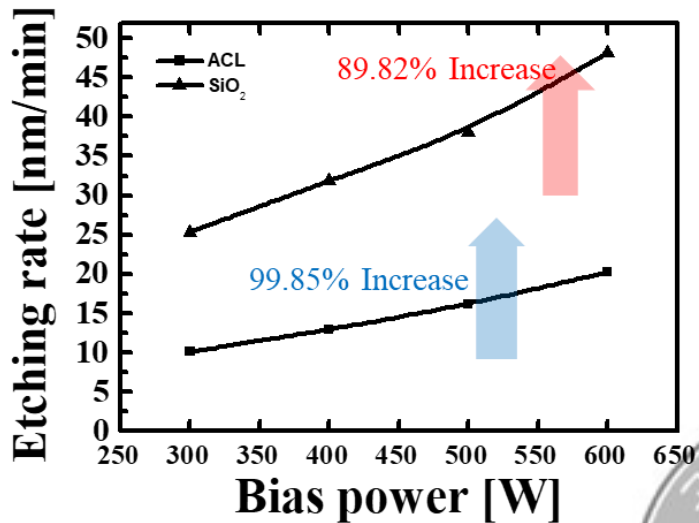
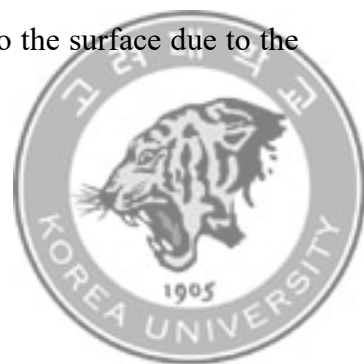


Figure 4.10 SiO₂, ACL etching rate according to the bias power increase



Figures 4.9 and 4.10 show the etch rates of SiO₂ and ACL under fixed conditions with an X gas fraction of 56.25% ($X = \text{C}_4\text{F}_6$, total flow rate: $\text{CF}_4 = 15$ sccm, $\text{C}_4\text{F}_6 = 25$ sccm, $\text{He} = 100$ sccm) as a function of source power and bias power. In the mixed plasma, as the source power increased from 50W to 200W and the bias power increased from 300W to 600W, the etch rates of both SiO₂ and ACL thin films monotonically increased. Specifically, as the source power varied from 50W to 200W, the SiO₂ etch rate increased from 24.89 to 36.15 nm/min (a 45.24% increase), and the ACL etch rate increased from 8.16 to 15.12 nm/min (an 85.29% increase). Similarly, as the bias power varied from 300W to 600W, the SiO₂ etch rate increased from 25.35 to 48.12 nm/min (an 89.82% increase), and the ACL etch rate increased from 10.12 to 20.225 nm/min (a 99.85% increase). This increase in etch rates is attributed to the higher ion flux and reactive neutral density generated by the increase in source power [38,83,84], and to the acceleration of ions towards the wafer with higher energy and momentum in a direction perpendicular to the surface due to the increase in bias power [38,84].



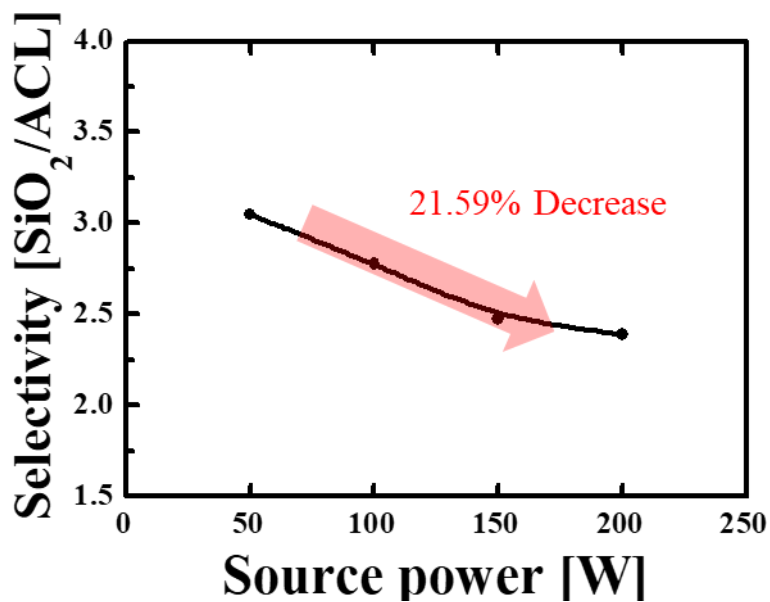


Figure 4.11 Selectivity (SiO_2/ACL) according to the Source power increase

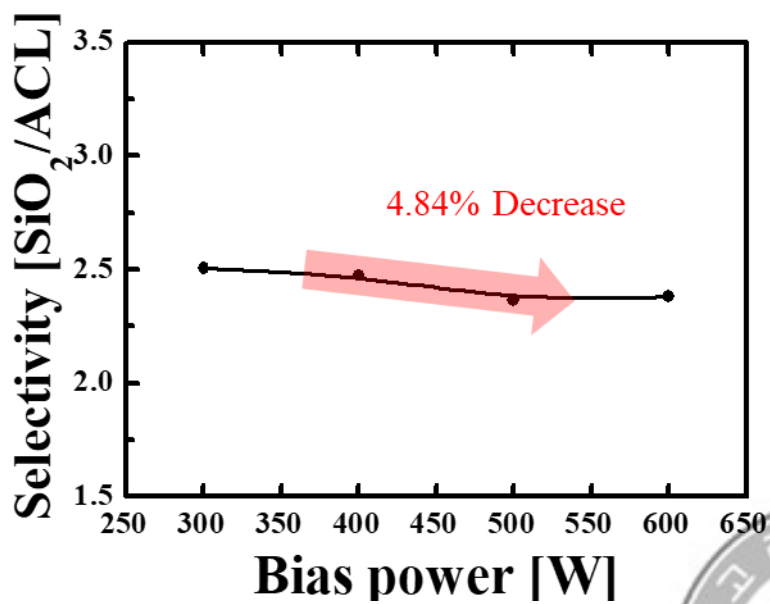
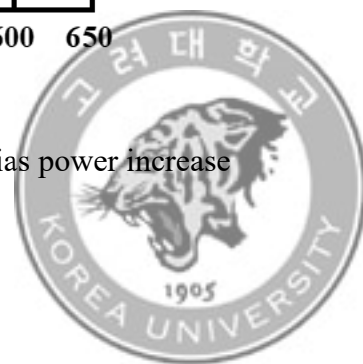


Figure 4.12 Selectivity (SiO_2/ACL) according to the Bias power increase



Conversely, although the etch rate increased with the increase in source power, the etch selectivity (SiO_2/ACL) decreased from 3.048 to 2.390 nm/min as the power increased, representing a 21.59% decrease. This is attributed to the higher increase rate of F radicals with the increase in source power, which leads to a reduction in the FC polymer participating in passivation [85-87]. With the increase in bias power, the etch rate also increased, but the etch selectivity (SiO_2/ACL) decreased from 2.50 to 2.379 nm/min, showing a smaller change (4.84% decrease). This is attributed to the increased damage from the higher ion bombardment energy [85-87].



4.2.2 Plasma diagnostic

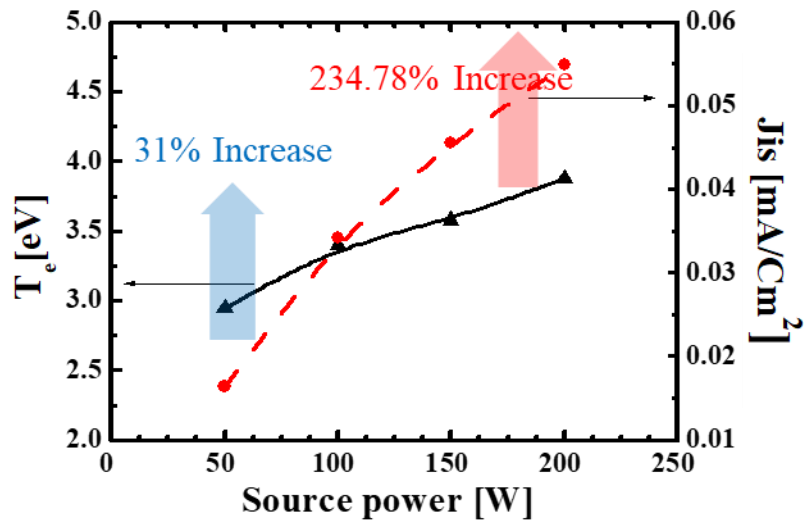


Figure 4.13 T_e & J_{is} according to Source power increase

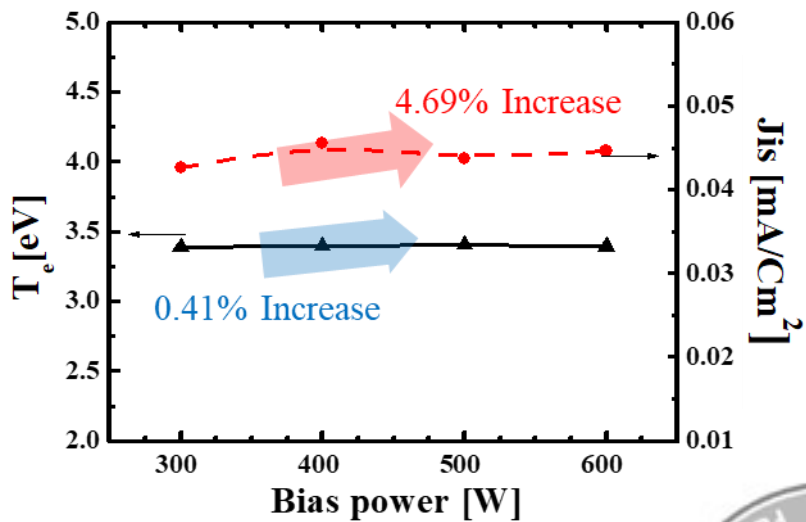


Figure 4.14 T_e & J_{is} according to Bias power increase



Figures 4.13 and 4.14 show the electron temperature and ion current density measured under fixed conditions with an X gas fraction of 56.25% ($X = \text{C}_4\text{F}_6$, total flow rate: $\text{CF}_4 = 15$ sccm, $\text{C}_4\text{F}_6 = 25$ sccm, $\text{He} = 100$ sccm) as the source power and bias power increase. With higher source power, the electron temperature and ion current density increased, with T_e changing from 2.94 to 3.87 eV (a 21% increase) and J_{is} changing from 0.01639 to 0.05487 mA/cm² (a 28% increase). The increase in electron temperature appears to be due to reduced electron energy losses from vibration and excitation of electrons. This effect enhances the electron-impact dissociation rate for multiple atomic components, enriching the gas phase with less saturated radicals and atomic species. The increase in ion current density seems to be due to the same change in total ionization rate accelerated by the growth of electron density. The relationship between bias voltage and electron density can be confirmed through the source power balance equation [88]. This phenomenon is commonly observed in many fluorocarbon-based plasma experiments under conventional reactive ion etching conditions [89-92]. When the bias power increased,



Te and Ji_s changed from 3.38 to 3.39 eV (a 4.69% increase) and from 0.042 to 0.044 mA/cm² (a 0.41% increase), respectively. It is widely known that in plasma, the frequency is divided into plasma ion frequency (low frequency) and plasma electron frequency (high frequency) [93-96]. Using a low-frequency bias (2 MHz), it was confirmed that bias power contributed minimally to plasma generation.

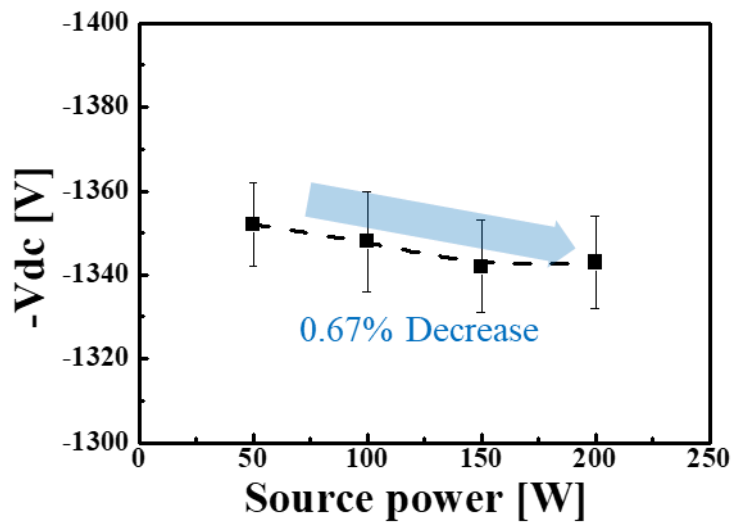


Figure 4.15 -Vdc according to Source power increase



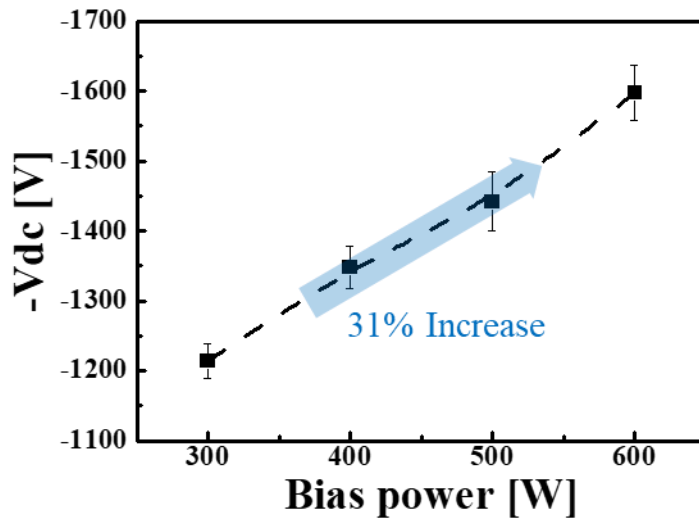
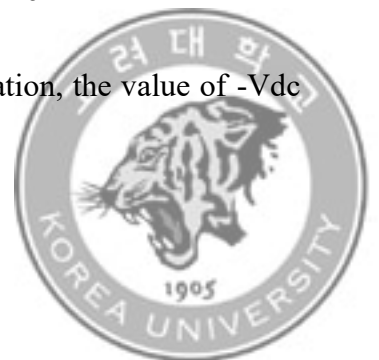


Figure 4.16 -Vdc according to Bias power increase

Figures 4.15 and 4.16 depict the variation of -Vdc with changes in source power and bias power. With an increase in source power, the value of -Vdc decreased slightly, showing minimal change (-1352 to -1343V, a 0.67% decrease). This slight decrease in -Vdc can be attributed to differences in plasma potential due to the increase in plasma density. Further investigation is needed to confirm the relationship between -Vdc and plasma density [84,97-99]. Additionally, Figure 4.16 illustrates that with the increase in bias power contributing to ion energy generation, the value of -Vdc



increased significantly (-1214 to -1598V, a 31% increase). This observation aligns with the analysis of etching characteristics, where an increase in bias power leads to an increase in etch rate. [100]



4.2.3 OES characteristics

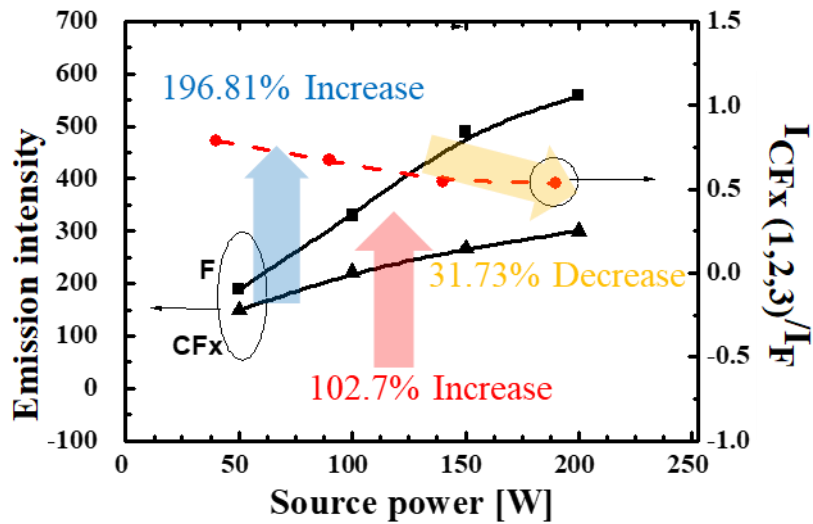


Figure 4.17 Emission intensity according to the Source power increase

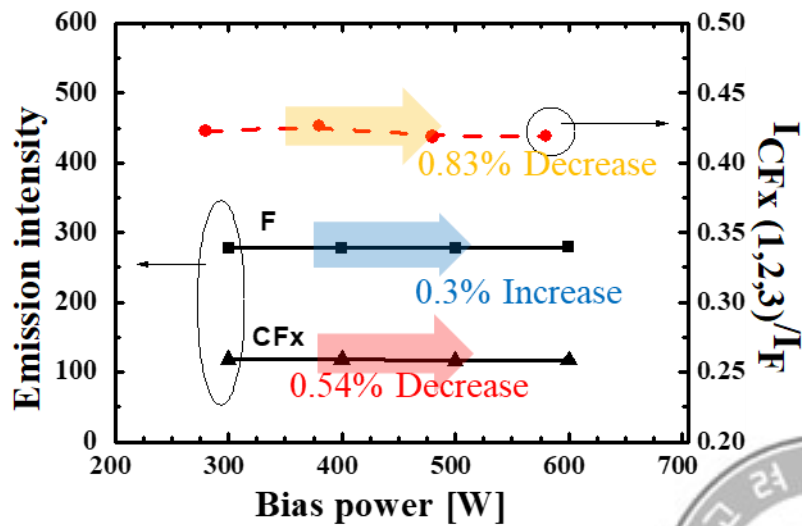
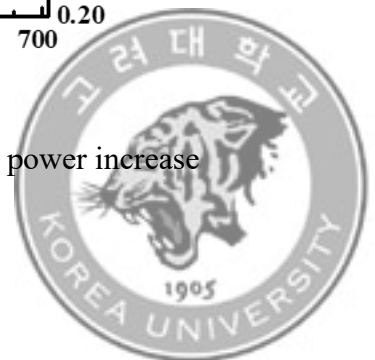
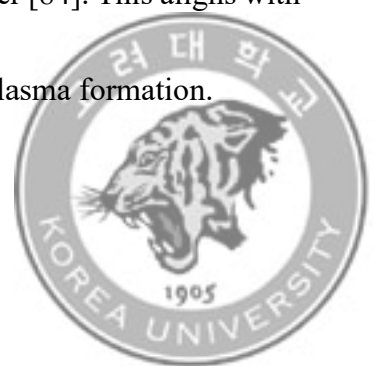


Figure 4.18 Emission intensity according to the Bias power increase



Figures 4.17 and 4.18 present optical emission intensity data obtained through OES to investigate the behavior of chemical reactive species such as polymers and fluorine under fixed conditions of X gas fraction 56.25% (X= C₄F₆, total flow rates CF₄ = 15sccm, C₄F₆ = 25sccm, He = 100sccm). Specific wavelengths of CF₁ (255.253nm), CF₂ (340.129 nm), CF₃ (610.144 nm), F (703.067 nm), and He (501 nm) were utilized [14,60,65]. In Figure 4.15, the intensity of each component increased with the increase in source power (CF_x = 148.95 – 300 (102.7% increase), F = 188.825 – 558.15 (196.81% increase)), indicating a reduction in electron temperature oscillation and electron energy loss due to the earlier explained increase in source power. [101,102]. This effect is attributed to the increase in electron-impact dissociation rate of the gas phase, enriched by less saturated radicals and atomic species, resulting in higher intensity peaks due to increased ionization. [103-105] For bias power, as reported, there was minimal variation in each peak with changes in bias power when using low-frequency bias in the ICP chamber [84]. This aligns with the inference that bias power has little influence on source plasma formation.



4.2.4 Etched profile

Figures 4.19 and 4.20 depict the etching profile images obtained from a $\text{CF}_4 + \text{C}_4\text{F}_6$ + He composite plasma as the source power and bias power increase ($\text{CF}_4 = 15\text{sccm}$ + $\text{C}_4\text{F}_6 = 25\text{sccm}$ + He = 100sccm Fixed). FE-SEM images of a 100nm line pattern with a 500nm width after plasma etching process are shown.

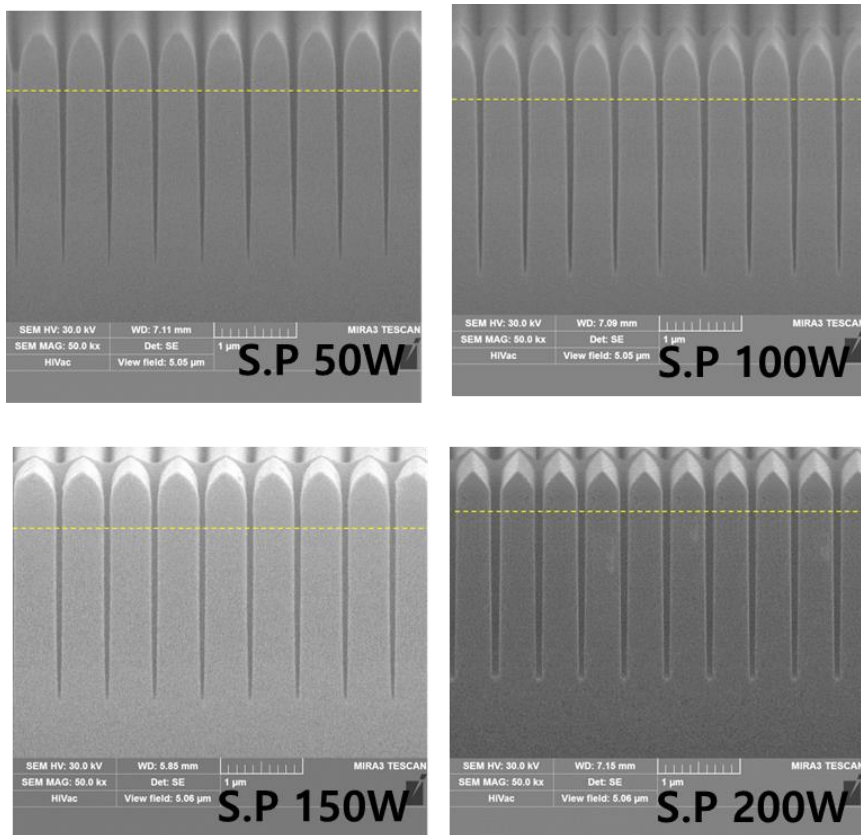


Figure 4.19 Etched profile in the Source power change



In Figure 4.19, SEM profile images are presented with varying source power from 50W to 200W. It is widely known in many fluorocarbon-based plasmas that increasing source power leads to higher ionization density due to increased ionization. [101,102]. As a result, the increase in dissociation of F atoms also leads to a reduction in passivation on the ACL mask surface, resulting in an increase in facet formation.

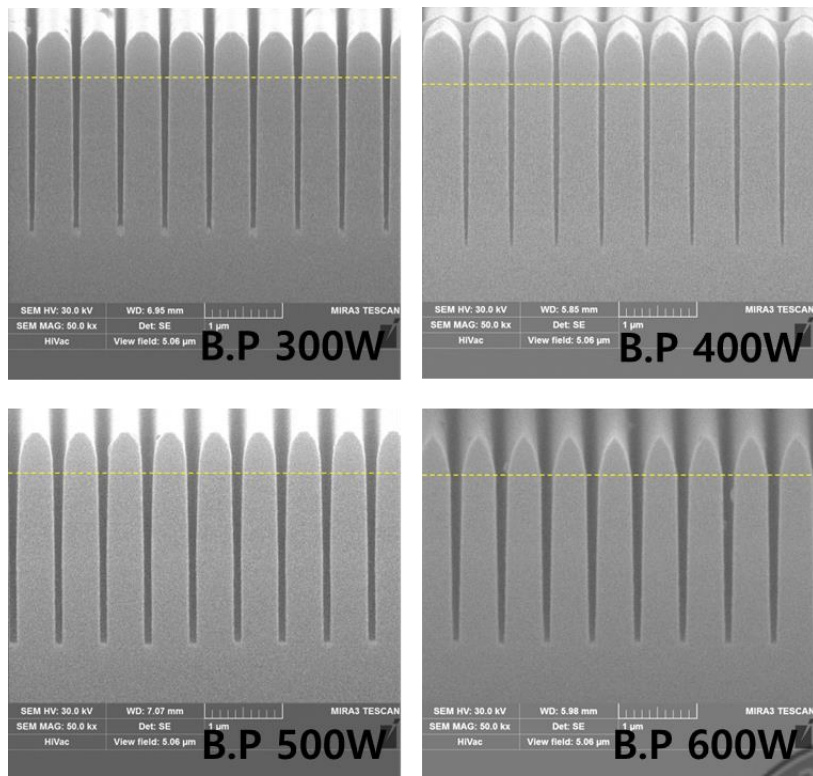


Figure 4.20 Etched profile in the Bias power change

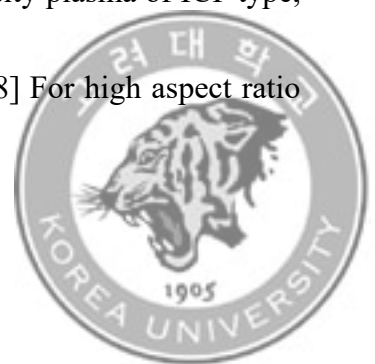


Figure 4.20 shows the SEM profile images as the Bias power is varied from 300W to 600W. The bias power applied to the lower electrode, contributing to the generation of $-V_{dc}$, increases the ion bombardment energy as the power increases. [106,107] Consequently, the increase in ion bombardment energy leads to damage on the Mask surface, resulting in an increase in ACL Mask erosion and facet formation. [106-108] This can be explained by the increase in $-V_{dc}$ observed in the plasma diagnostics conducted earlier with increasing bias power.



5. Conclusion

In this study, plasma etching characteristics and plasma diagnostics were conducted by varying three parameters: gas ratio, Source power, and Bias power in a $\text{CF}_4 + \text{C}_4\text{F}_6 + \text{He}$ mixed plasma. With the increasing complexity of circuit structures and the repetition of processes in multiple-patterning techniques, it has been reported that the flattening of etching profiles on both mask and pattern bottom surfaces is crucial to prevent misalignment during patterning. [109,110] Additionally, due to the complexity of circuits, efforts have been made to reduce carbon emissions and explore new materials and gases. [111] C_4F_6 is an environmentally friendly gas with a GWP index of less than 1 and a lifetime of 2 days. Recent research on high aspect ratio etching using C_4F_6 gas and C_4F_8 gas has regained attention. However, previous studies only focused on basic etching evaluation and plasma diagnostics using these gases. [56-58] There were no studies specifically on high aspect ratio etching. Furthermore, it has been reported that research on high-density plasma of ICP type, as opposed to the conventional CCP type, is needed. [27-28] For high aspect ratio

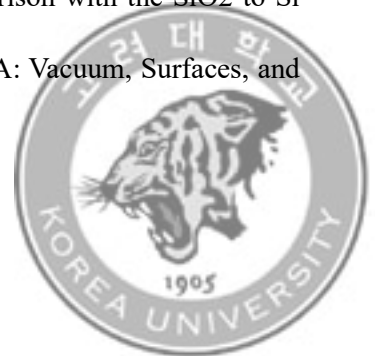


etching, research on process optimization for conventional gases is necessary in addition to exploring new variables. We investigated the potential applicability of C_4F_6 as an environmentally friendly alternative gas for high aspect ratio oxide etching and examined the etching characteristics and plasma properties observed when using C_4F_6 plasma with various parameter variations. Additionally, the most vertical profiles were observed under the condition of $CF_4 / C_4F_6 / He$ (15/25/100, sccm). In this study, by finely adjusting each parameter, we proposed the possibility of HAR etching through process optimization.

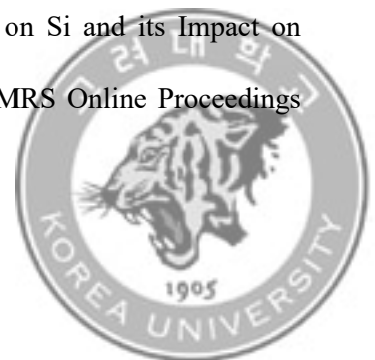


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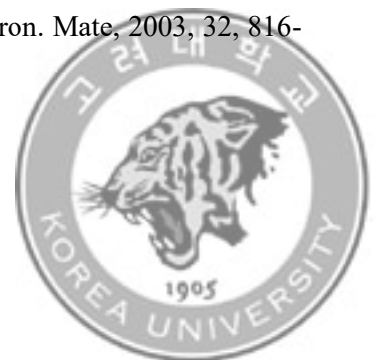
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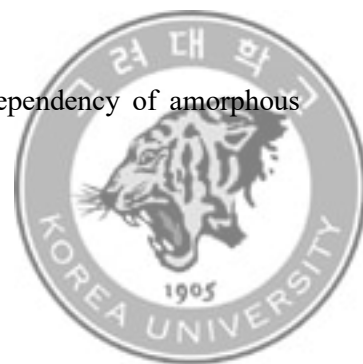
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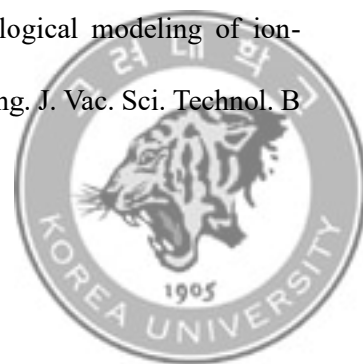


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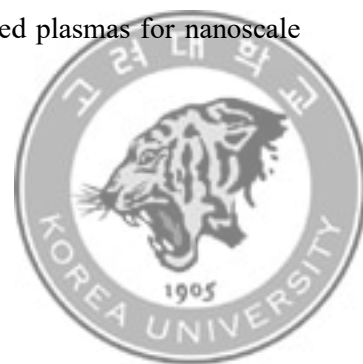
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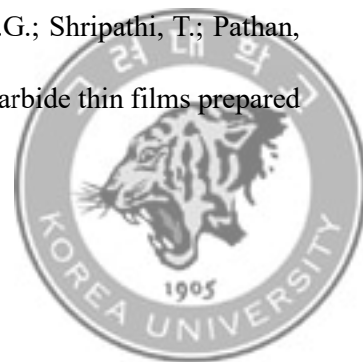


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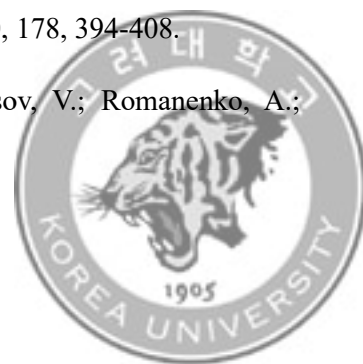


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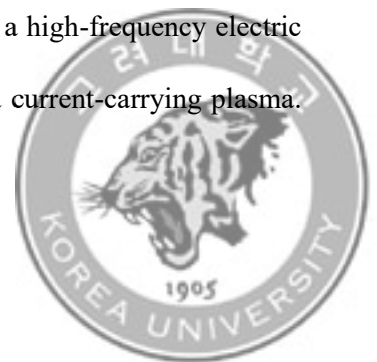
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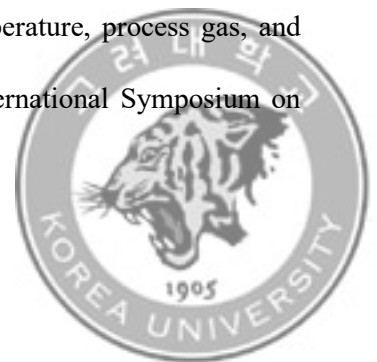


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